Tomita–Takesaki modular structure on truncated SU(2) spectral triples: the four-witness 2π period at finite cutoff

J. Loutey*

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Abstract

The four-witness Wick-rotation theorem (Hartle–Hawking [13], Sewell [24], Bisognano–Wichmann [2], Unruh [29]) asserts that a state of a quantum field restricted to a region bounded by a bifurcate Killing horizon with surface gravity κ_g is KMS at inverse temperature $\beta = 2\pi/\kappa_g$ with respect to Killing-time evolution. The shared 2π across all four faces is $\mathrm{Vol}(S^1)$, the regularity period of the Euclidean rotation of the observer's causal-domain coordinates. That continuum statement is prior art and is not re-proved here. We study its finite-cutoff *period* on the Connes–van Suijlekom [10] truncated Camporesi–Higuchi [4] spectral triple $\mathcal{T}_{n_{\max}}$, with a hemispheric wedge and a posited wedge generator $K_\alpha^W = \mathrm{diag}(2m_j)$ on the $m_j > 0$ wedge states, a relabelling built from the doubled polar rotation generator $2J_z$ (whose own compression to the wedge is zero). Because K_α^W has integer spectrum, $e^{2\pi i K_\alpha^W} = I$, and the conjugation flow closes, $\sigma_{2\pi}(O) = O$, for *every* operator O — an elementary fact [SYMBOLIC] that holds for any integer-spectrum generator whatsoever and therefore does not single out a Bisognano–Wichmann modular flow. The Tomita–Takesaki reading is not an independent check: the wedge state is *defined* as $\rho_W = e^{-K_\alpha^W}/Z$, so Tomita's theorem for a Gibbs state returns the same generator, $\sigma_t^{\mathrm{TT}} = \sigma_{-t}^{\alpha}$. Nor is the agreement of the four witnesses (Bisognano–Wichmann, Hartle–Hawking, Sewell, Unruh) a finite-cutoff result: the six witness instantiations use the same K_α^W with the surface gravity κ_g relabelled, and $\beta H_{\mathrm{local}} = K_\alpha^W$ is κ_g-independent by construction, so their bit-identical residuals hold by construction. We record these closures (bit-exact in floating point at $n_{\max} \in \{2,3,4,5\}$, residual $O(\sqrt{\dim \mathcal{H}_{n_{\max}}} \cdot \varepsilon_{\mathrm{machine}})$) as what they are. We also record the operator-level distinction between $2J_z$ ($e^{2\pi i K} = +I$) and the wedge-restricted Camporesi–Higuchi Dirac (half-integer spectrum, $e^{2\pi i D_W} = -I$, the spinor double-cover sign; the conjugation flow closes for both). The genuine finite-cutoff question — *derive* K as the modular Hamiltonian of a wedge-restricted Camporesi–Higuchi state, rather than posit it — is [OPEN]. The Lorentzian extension to signature $(3,1)$ via Bizi–Brouder–Besnard 2018 [3] Krein-space spectral triples is carried out *at finite cutoff* in the present paper (§11, which absorbs the former Paper 43): a Krein carrier and Krein-self-adjoint Lorentzian Dirac, an axiom audit at Bizi–Brouder–Besnard signature $(4,6)$ in which the universal chirality anticommutation fails on the truthful Camporesi–Higuchi Dirac, the same automatic (and signature-blind) period closure, and a Hilbert–Schmidt orthogonality between the posited local Hamiltonian and the wedge Dirac that is a chirality-parity identity [SYMBOLIC], the only transcendental of its residual, $1/\pi^2$, being the posited $\beta = 2\pi/\kappa_g$ squared. Its *continuum* (Lorentzian-propinquity-rate) closure was described here as the named open frontier when this paper was drafted; that is **superseded** (2026-06, Paper 45 [44]): the natural device fails structurally, and the chase is closed on the convention branch (an [ARGUED] reading: its Euclidean-only leg is generator-independent, its Wick-rotation/KMS-circle leg is conditional on O0 for the Bisognano–Wichmann generator; Paper 45 Q1). The mechanism

*Independent researcher. jloutey@gmail.com. This work was developed in an AI-augmented agentic workflow with Anthropic's Claude. Implementation, internal memos, and bibliographic collation were performed collaboratively; all theorems, design choices, and editorial decisions are the author's responsibility.

is visible in this paper's own result (i): the posited wedge generator has *integer* spectrum, so the finite-cutoff flow is a compact $\beta = 2\pi$ circle rather than a non-compact boost [SYMBOLIC], and on the argued reading [ARGUED] every Lipschitz/boost seminorm built on it is signature-blind. The finite-cutoff closures reported below stand as computed; what does not follow from them is a metric statement about *signature*, or an identification of the posited generator with a Bisognano–Wichmann boost. The construction is a finite-dimensional illustration of the Connes–Rovelli thermal-time setup [9] on a hemispheric wedge of the round S^3, not a derivation of it.

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Keywords: Tomita–Takesaki modular flow, Bisognano–Wichmann theorem, Connes–van Suijlekom truncated operator system, Camporesi–Higuchi Dirac, four-witness Wick rotation, Hartle–Hawking, Sewell, Unruh effect, KMS state, SU(2) spectral triple.

1 Introduction

A central thread of axiomatic quantum field theory is the *four-witness Wick-rotation theorem*: under suitable analytic hypotheses, a state of a quantum field restricted to a region bounded by a bifurcate Killing horizon with surface gravity κ_g is a KMS state at inverse temperature $\beta = 2\pi/\kappa_g$ with respect to evolution along the relevant Killing vector field. The four physical instantiations are Bisognano–Wichmann [2] for the Rindler wedge of Minkowski spacetime, Hartle–Hawking [13] for the Hartle–Hawking thermal vacuum on the Schwarzschild exterior, Sewell [24] for the abstract KMS-modular reading of the Hartle–Hawking vacuum on globally hyperbolic manifolds with bifurcate Killing horizons, and Unruh [29] for the proper-acceleration parameterisation of the Bisognano–Wichmann setup. The shared 2π across all four faces is $\mathrm{Vol}(S^1)$, the regularity period of the Euclidean rotation of the observer's causal-domain coordinates after Wick rotation $\tau = it$.

The theorem closes naturally in axiomatic Wightman QFT for free fields on globally hyperbolic Lorentzian spacetimes. Its operator-system analog on a finite-dimensional truncation of a spectral triple — i.e., the question whether the modular automorphism associated to a wedge-restricted Gibbs state on a finite-dimensional operator system realises the same $\sigma_{i\cdot 2\pi} =$ identity period closure intrinsically, without recourse to the Wightman axiomatic Wick-rotation chain — is structurally distinct, and has not been settled in the spectral-truncation literature of [10, 19].

The present paper does *not* settle that question. It sets up the finite-cutoff ingredients (wedge, Gibbs state, GNS/Tomita construction) on the truncated Camporesi–Higuchi spectral triple $\mathcal{T}_{n_{\max}}$ over the round $S^3 = \mathrm{SU}(2)$, records what holds when the modular generator is *posited* to be the doubled rotation generator $2J_z$, and isolates what remains open: deriving the generator from a wedge-restricted Camporesi–Higuchi state. We work in the setup of Connes–van Suijlekom [10], with the truncated operator system $\mathcal{O}_{n_{\max}} = P_{n_{\max}}\, C^\infty(S^3)\, P_{n_{\max}}$ obtained by orthogonal projection onto the finite-dimensional subspace $\mathcal{H}_{n_{\max}}$ spanned by the lowest Camporesi–Higuchi spinor harmonics, and with the Connes–vS "Toeplitz S^1" propagation invariant $\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2$ verified verbatim in supporting work of the present author (Paper 32 § III). The convergence of $\mathcal{T}_{n_{\max}}$ to the round-S^3 Camporesi–Higuchi spectral triple $\mathcal{T}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance — with explicit rate $d_{\mathrm{GH}}(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \le \gamma_{n_{\max}}$ where $\gamma_{n_{\max}} \sim (4/\pi)\log n_{\max}/n_{\max}$, for the commutative factor on the true window — has been established in the present author's Paper 38 [40].

Theorem 1.1 (Main result, [SYMBOLIC]; automatic for any integer-spectrum generator; cf. Section 7)**.** *Let $\mathcal{T}_{n_{\max}}$ denote the truncated Camporesi–Higuchi spectral triple of cutoff $n_{\max}$ over the*

round $S^3 = \mathrm{SU}(2)$, let P_W denote the hemispheric wedge projector defined by the equatorial reflection $R_{\mathrm{polar}} : m_j \mapsto -m_j$ on the full-Dirac spinor basis, and let $\omega_W^{\mathrm{BW}}(O) = Z^{-1}\mathrm{Tr}_{P_W\mathcal{H}_{n_{\max}}}(e^{-K_\alpha^W} O)$ denote the wedge-restricted Gibbs state at the BW local Hamiltonian $H_{\mathrm{local}} = K_\alpha^W/\beta$. Then for every $n_{\max} \geq 1$, both the geometric BW-α modular flow $\sigma_t^\alpha(O) = e^{itK_\alpha^W} O e^{-itK_\alpha^W}$ and the Tomita–Takesaki BW-γ modular flow $\sigma_t^{\mathrm{TT}}(O) = \Delta^{it} O \Delta^{-it}$, with $\Delta = S^*S$ extracted from the polar decomposition $S = J_{\mathrm{TT}}\Delta^{1/2}$ on the GNS Hilbert–Schmidt space, satisfy the period closure

$$\sigma_{2\pi}^\alpha(O) \;=\; O, \qquad \sigma_{2\pi}^{\mathrm{TT}}(O) \;=\; O \tag{1}$$

bit-exactly for every $O \in P_W\mathcal{O}_{n_{\max}}P_W$, with maximum periodicity residual scaling as $O(\sqrt{\dim \mathcal{H}_{n_{\max}}} \cdot \varepsilon_{\mathrm{machine}})$. Furthermore, the two flows are operator-action conjugates,

$$\sigma_t^{\mathrm{TT}}(O) \;=\; \sigma_{-t}^\alpha(O) \tag{2}$$

bit-exact at every real t, and the density matrix $\rho_W = e^{-K_\alpha^W}/Z$ is β-independent, so the six witness instantiations $(\mathrm{BW}, \mathrm{HH}_{M=1}, \mathrm{HH}_{M=2}, \mathrm{Sew}_{M=1}, \mathrm{Unruh}_{a=1}, \mathrm{Unruh}_{a=2})$ produce identical Δ and K_{TT}.

Every clause of Theorem 1.1 is elementary, and none of it is evidence for a Bisognano–Wichmann identification at finite cutoff. (i) The period closure holds for *any* Hermitian generator with integer spectrum and *any* operator O: $e^{2\pi i n} = 1$ for integer n. Replacing $2J_z$ by an unrelated integer-spectrum matrix leaves every closure test green. (ii) The conjugacy $\sigma_t^{\mathrm{TT}} = \sigma_{-t}^\alpha$ is Tomita's theorem for a Gibbs state $\rho = e^{-K}/Z$, which returns K; since the state was built from K_α^W, the Tomita construction is circular as a check on K_α^W. (iii) The witness agreement is by construction: the six instantiations share K_α^W and differ only in the label κ_g, which cancels in βH_{local}. The generator $K_\alpha^W = \mathrm{two_}m_j$ (the positive odd integers $\leq 2l+1$ on the wedge states) is a relabelling built from the polar *rotation* generator on the Riemannian S^3, chosen, not derived, to play the role of the boost. The continuum four-witness statement ($\beta = 2\pi/\kappa_g$; [2, 13, 24, 29]) is prior art and is unaffected; its finite-cutoff counterpart — obtaining K as the modular Hamiltonian of a wedge-restricted Camporesi–Higuchi state and *then* checking the period — is [OPEN] (Section 10, O0).

What this paper does and does not close. This paper does *not* close the principal Sprint L1[1] falsifier named in Paper 32 § VIII Remark `rem:bisognano_wichmann_reading` of the present author's framework documentation: "compute the modular Hamiltonian K on $\mathcal{T}_{n_{\max}}$ for a wedge-restricted Camporesi–Higuchi state and verify $\sigma_{i\cdot 2\pi}$ = identity at finite $n_{\max}$." That falsifier asks for K to be *computed from* a Camporesi–Higuchi state; here K is posited ($2J_z$) and the state is built from it, so the period closure is automatic and tests nothing about the state. The supporting module reports the labels STRONG_IDENTIFICATION (BW-α) and UNIFIED_STRONG (BW-γ); they record that the automatic closure was reproduced numerically, and are *not* an identification verdict. The falsifier remains [OPEN]. The Lorentzian extension to signature $(3,1)$ via the Bizi–Brouder–Besnard 2018 [3] Krein-space spectral triple is carried out *at finite cutoff* in §11 (the same automatic period closure $\sigma_{2\pi}$ = id on the Krein wedge, not a four-witness identification); the continuum / Lorentzian-propinquity-rate extension is *not* closed by this paper (the named open target, and Paper 45's degeneracy theorem closes off the natural Lorentzian-propinquity route). Sprint L0 of the present author's framework documentation (`debug/lorentzian_l0_audit_memo.md`, Paper 31 § 8, Paper 34 § V.E) audits the 28 projections of the framework's projection taxonomy and predicts that the M3 sub-mechanism of the master Mellin engine becomes trivially empty at signature $(3,1)$ on chirality-symmetric spectrum, leaving the M1 $\mathrm{Vol}(S^1) = 2\pi$ mechanism (the load-bearing engine for the four-witness theorem) intact under signature change; that prediction turns out to be convention-dependent

[1] Sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository).

(§11.8). The automatic closure of the present paper does not confirm an operator-system identification at signature $(3,0)$; it shows only that the posited generator has the integer spectrum a 2π period requires.

Related and concurrent work. The most directly comparable construction in the published literature is the Connes–Rovelli [9] thermal time hypothesis, which identifies the Tomita modular flow on a covariant quantum field theory as the physical time evolution. The Connes–Rovelli identification is general-covariant and abstract; the present paper specialises it to a finite-dimensional operator system in the Connes–van Suijlekom [10] truncation framework on the round S^3. The continuum side has the long tradition starting from Bisognano–Wichmann [2] and standardised through Casini–Huerta [5] and Casini–Huerta–Myers [6], with the continuum modular Hamiltonian for a hemisphere of S^d in CFT given explicitly as $K_{\text{cont}} = 2\pi \int_{H_+} \xi \cdot T^{00}\, d\Omega$ where ξ is the wedge-preserving Killing field. The present finite-cutoff construction mimics only the *form* of this identification: the integer spectrum of the posited generator stands in for the canonical 2π-period of the continuum modular flow, but the generator is not obtained from a state. A finite-lattice realisation in critical quantum spin chains was given by Zhang *et al.* [32], where the BW form is *extracted* from a ground state and so is a genuine test; the present construction does not attempt that extraction, and differs also in that the underlying object is a continuum spectral triple on a non-flat compact S^3 background, and the finite-dimensional structure comes from the spectral truncation of D_{CH} rather than from a discretisation of physical space. The existing modular literature on Riemannian backgrounds includes Strohmaier [25] (Reeh–Schlieder on stationary spacetimes) and Verch [31] (generally covariant spin-statistics and modular structure on curved backgrounds); these address the algebraic-QFT side, whereas the present paper anchors the modular structure on the Connes–van Suijlekom truncated operator system. Direct attempts to lift the Wick-rotation chain at the spectral-triple level via twisted-spectral-triple morphisms have been pursued recently by Nieuviarts [21–23], but the twist construction (Definition 2.2 of [22]) is explicitly restricted to even-dimensional manifolds and does not apply to the S^3 case treated here.

Outline. Section 2 recalls the truncated Camporesi–Higuchi spectral triple and the Paper 38 state-space Gromov–Hausdorff convergence result. Section 3 summarises the Sprint L0 Lorentzian-readiness audit, names the Riemannian scope of the present paper, and footnotes the Nieuviarts NO-GO on direct twist-morphism emergence for odd-dim S^3. Section 4 sets up the hemispheric wedge P_W, the KMS state at $\beta = 2\pi$ on the wedge sub-algebra, and the unified surface-gravity normalisation. Section 5 constructs the posited geometric BW-α generator $K_\alpha = J_{\text{polar}}$ (its identification as a modular Hamiltonian is open, O0) and proves the automatic period closure Theorem 5.4. Section 6 constructs the Tomita–Takesaki BW-γ modular Hamiltonian $K_{\text{TT}} = -\log \Delta$ of the Gibbs state built from that posited generator, via polar decomposition on the GNS Hilbert–Schmidt space, distinguishes J_{TT} ($J^2 = +I$) from the Connes J_{GV} ($J^2 = -I$), and proves Theorem 6.3 (the same closure, read through the Gibbs state built from the posited generator). Section 7 proves Theorem 7.1 (the flow conjugacy $\sigma_t^{\text{TT}} = \sigma_{-t}^{\alpha}$, Tomita's theorem for a Gibbs state) and discusses the local Hamiltonian $H_{\text{local}} = K_\alpha^W/\beta$ as a posited choice. Section 8 records that the HH, Sewell, BW, and Unruh instantiations coincide by construction. Section 9 places the result in the modular-theory and NCG literature. Section 10 summarises and names open questions. Section 11 carries the construction onto a signature-$(3,1)$ Krein carrier at finite cutoff (the successor of the former Paper 43). Appendix A documents the computational verification protocol and the supporting module `geovac/modular_hamiltonian.py`. Appendix B cross-references the result to the present author's broader framework (Paper 38 [40] for state-space Gromov–Hausdorff convergence; Paper 40 [42] for the $4/\pi$ Fejér-moment constant reported universal over compact simple groups (derived at rank 1); Paper 32 § VIII for the master Mellin engine case-exhaustion theorem and the four-witness theorem's M1 sub-mechanism reading).

2 The truncated Camporesi–Higuchi spectral triple

We work in the framework of metric spectral triples and their spectral truncations established by Connes [8, 10] and developed in the GH-convergence context by Latrémolière [17, 18], Hekkelman [14], Leimbach–van Suijlekom [19], and the present author's Paper 38 [40] for the round $S^3 = \mathrm{SU}(2)$ case.

2.1 The continuum Camporesi–Higuchi triple

Identify the unit round S^3 with the Lie group SU(2) via the standard double cover, equipped with the bi-invariant metric of total volume $\mathrm{Vol}(S^3) = 2\pi^2$. Let Σ denote the spinor bundle and $L^2(S^3, \Sigma)$ the spinor Hilbert space. The *Camporesi–Higuchi Dirac operator* $D_{\mathrm{CH}} : \mathrm{Dom}(D_{\mathrm{CH}}) \subset L^2(S^3, \Sigma) \to L^2(S^3, \Sigma)$ acts diagonally on the spinor harmonic basis $\{\psi_{n,l,m_j,\chi}\}$ via

$$D_{\mathrm{CH}}\, \psi_{n,l,m_j,\chi} \;=\; \chi\,(n + \tfrac{1}{2})\, \psi_{n,l,m_j,\chi}, \tag{3}$$

with $n \in \mathbb{N}_{\geq 1}$, $l \in \{0, 1, \ldots, n-1\}$, $j = l + 1/2$, $m_j \in \{-j, -j+1, \ldots, j\}$, and chirality $\chi \in \{+1, -1\}$ [4]. The degeneracy of the level n on each chirality sector is $n(n+1)$; the total degeneracy including both chiralities is $2n(n+1)$.

We form the spectral triple

$$\mathcal{T}_{S^3} \;=\; \big(C^\infty(S^3),\; L^2(S^3, \Sigma),\; D_{\mathrm{CH}}\big). \tag{4}$$

This is a unital, even, real spectral triple of metric dimension 3 in the sense of Connes [8], with KO-dimension 3 (mod 8). The chirality grading $\gamma_5 = \mathrm{diag}(+1, -1)$ on $\Sigma_+ \oplus \Sigma_-$, the Connes real structure J_{GV} (charge conjugation, $J_{\mathrm{GV}}^2 = -I$), and the standard Lipschitz seminorm $\|[D_{\mathrm{CH}}, M_f]\|_{\mathrm{op}}$ on the algebra $C^\infty(S^3)$ all enter as expected. We use the labelling $n_{\mathrm{Fock}} = n_{\mathrm{CH}} + 1$ throughout, so the lowest non-trivial level is $n = 1$ with $|\lambda_1| = 3/2$.

2.2 Truncations and the operator system

Let $P_{n_{\max}} : L^2(S^3, \Sigma) \to \mathcal{H}_{n_{\max}}$ denote the orthogonal projection onto the finite-dimensional subspace spanned by the spinor harmonics with $1 \leq n \leq n_{\max}$. The truncated Hilbert space has dimension

$$\dim \mathcal{H}_{n_{\max}} \;=\; 2 \sum_{n=1}^{n_{\max}} n(n+1) \;=\; \tfrac{2}{3}\, n_{\max}(n_{\max}+1)(n_{\max}+2). \tag{5}$$

Numerically: $\dim \mathcal{H}_2 = 16$, $\dim \mathcal{H}_3 = 40$, $\dim \mathcal{H}_4 = 80$, $\dim \mathcal{H}_5 = 140$. The $n_{\max} = 3$ case is numerically distinguished by the coincidence $\dim \mathcal{H}_3 = 40 = g_3^{\mathrm{Dirac}} = \Delta^{-1}$ — different objects: $\dim \mathcal{H}_3 = 4 + 12 + 24$ is the cumulative node count over the Camporesi–Higuchi levels $k \leq 2$, while $g_3^{\mathrm{Dirac}} = 2(3+1)(3+2) = 40$ is the both-sign Dirac degeneracy of the single level $k = 3$ ($|\lambda| = 9/2$); they agree only at $n_{\max} = 3$ — where Δ^{-1} is the boundary-mass invariant of the present author's Paper 2 [33] fine-structure-constant observation.

The truncated Dirac operator is $D_{n_{\max}} = P_{n_{\max}} D_{\mathrm{CH}} P_{n_{\max}}$, which (since D_{CH} is diagonal in the spinor basis) acts as $\chi\,(n + 1/2)$ on each kept basis vector. Following Connes–van Suijlekom [10], the *truncated operator system* is

$$\mathcal{O}_{n_{\max}} \;=\; P_{n_{\max}}\, C^\infty(S^3)\, P_{n_{\max}} \;\subset\; B(\mathcal{H}_{n_{\max}}), \tag{6}$$

viewed as a self-adjoint subspace of the matrix algebra $M_{\dim \mathcal{H}_{n_{\max}}}(\mathbb{C})$. As emphasised in [10, §1], this is *not* a C^*-subalgebra of $M_{\dim \mathcal{H}_{n_{\max}}}(\mathbb{C})$: the full C^*-algebra it generates is the full matrix algebra (the "C^*-envelope"), but the operator system $\mathcal{O}_{n_{\max}}$ itself encodes finer structure, including the algebra-side analog of the Lipschitz seminorm and the propagation invariant. The truncated metric spectral triple is then $\mathcal{T}_{n_{\max}} = (\mathcal{O}_{n_{\max}}, \mathcal{H}_{n_{\max}}, D_{n_{\max}})$.

Paper 32 § III of the present author's framework documentation verifies that the propagation invariant $\mathrm{prop}(\mathcal{O}_{n_{\max}}) = 2$ at $n_{\max} \in \{2, 3, 4\}$, matching Connes–van Suijlekom [10, §4] for the Toeplitz operator system on S^1 verbatim. This is the structural match that establishes $\mathcal{O}_{n_{\max}}$ as the right operator-system host for spectral-triple-internal modular constructions.

2.3 State-space GH convergence to the continuum

The present author's Paper 38 [40] proves that the truncated triples $\mathcal{T}_{n_{\max}}$ converge to $\mathcal{T}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance with explicit rate

$$d_{\mathrm{GH}}(\mathcal{T}_{n_{\max}}, \mathcal{T}_{S^3}) \;\le\; \gamma_{n_{\max}}, \qquad \gamma_{n_{\max}} \;=\; \frac{4 \log n_{\max}}{\pi\, n_{\max}} \;+\; O(1/n_{\max}), \tag{7}$$

(commutative factor, true window) with Fejér-moment constant $4/\pi = \mathrm{Vol}(S^2)/\pi^2$ (in the dual-Coxeter normalisation of the SU(2) metric; $2/\pi$ on the unit round S^3, the constant scaling with the metric; a tapered amplitude gives $d_{\mathrm{GH}} = O(1/n_{\max})$ **[MEASURED]**, Paper 38). It is read (an Observation-level reading, not a derivation) as carrying the $k = 0$ volume content of the M1 sub-mechanism of the master Mellin engine. It is *not* the Hopf fibration's base-to-total ratio and not an $\mathrm{SU}(2)/U(1)$ Haar quotient: the Hopf map is a Riemannian submersion $S^3(r) \to S^2(r/2)$, whose ratio is neither $4/\pi$ nor $2/\pi$; the constant is twice the quotient of *unit-radius* sphere volumes, $\mathrm{Vol}(S^2)/\mathrm{Vol}(S^3) = 2/\pi$. (The $k = 0$ Mellin slot is a volume slot; Paper 25's $\mathrm{Vol}(S^2)/4 = \pi$ is a quarter of the unit-sphere area $\mathrm{Vol}(S^2) = 4\pi$ (equivalently the full area of the Hopf base $S^2(1/2)$ of the unit S^3). Paper 38, volume-ratio note.) The M1 reading is on S^3 (the Paper 32 § VIII case-exhaustion, **[CONDITIONAL]**, does not itself cover GH rates; the Fejér-moment bound is generalised to compact connected semisimple Lie groups in Paper 40 [42], for general G conditionally on the per-group spin-window decomposition; a circle factor carries the circle constant). The lemmas L1' / L2 / L3 / L4 of Paper 38 — chirality-doubled operator system substrate, central spectral Fejér kernel on SU(2), Lipschitz comparison bound, Berezin reconstruction — supply the analytic infrastructure that the present paper builds on. (Paper 38's fifth lemma, the Latrémolière propinquity assembly, was *withdrawn* on 2026-09-03 when its height leg was refuted; nothing here depends on it, and Paper 38's convergence theorem is carried by the lifted-state construction instead.)

The present paper does not require the GH-convergence machinery in any load-bearing way: the period closure proved below is a finite-cutoff bit-exact identity, not a limit statement. The state-space Gromov–Hausdorff convergence (7) enters only as a *consistency* sanity check (Section 6); it does not link the modular period to the metric-side Fejér-moment constant.

3 Lorentzian-readiness audit and Riemannian scope

The setting of the present paper is signature $(3, 0)$: the round S^3 Camporesi–Higuchi spectral triple is Riemannian, with KO-dimension 3 (mod 8). The four-witness Wick-rotation theorem inhabits a hybrid status: each of the four physical witnesses (Hartle–Hawking, Sewell, Bisognano–Wichmann, Unruh) is naturally a *Lorentzian* statement at signature $(3, 1)$ in the Wightman axiomatic framework, but the shared $\beta = 2\pi/\kappa_g$ structural identity that unifies the four faces lives entirely on the Euclidean side via Wick rotation. The Wick-rotation chain itself [Hartle–Hawking 1976 → Sewell 1982 → Bisognano–Wichmann 1976 → Unruh 1976] operates at the metric-functional level: it converts a Euclidean heat-kernel-style spectral computation into its Lorentzian modular-flow interpretation without re-deriving the underlying spectral object in Krein signature.

3.1 The 28-projection Lorentzian transfer audit

The present author's Sprint L0 audit (`debug/lorentzian_l0_audit_memo.md`, Paper 31 § 8, Paper 34 § V.E) classifies the 28 projections of the framework's projection taxonomy into four buckets:

(T) *TRANSFERS_FREELY:* signature-blind, lives on the variable/dimension axes of the projection taxonomy only, with no Riemannian volume form, no Mellin transform, and no Latrémolière propinquity. Holds at $(3,0)$ and $(3,1)$ identically.

(W) *WICK_MAP_FREE:* reduces to the M1 sub-mechanism of the master Mellin engine via the four-witness Wick-rotation theorem at the metric-functional level.

(E) *EUCLIDEAN_SPECIFIC:* requires either (i) a Krein-space spectral triple at $(3,1)$ per Bizi–Brouder–Besnard 2018 [3], (ii) a Lorentzian-native heat-trace formalism, or (iii) a Lorentzian analog of the Latrémolière propinquity (which does not yet exist in the published literature).

(M) *MIXED:* parts transfer freely, parts require extension.

The headline distribution at May 2026 is $17/4/5/2 = 61\%/14\%/18\%/7\%$: three-quarters of the projection dictionary inherits a Lorentzian reading at no further structural cost at the metric-functional level. The Wick-rotation projection itself sits in the (W) bucket and carries the M1 transcendental signature $2\pi \cdot \mathbb{Q}$ via $\mathrm{Vol}(S^1)$. The five EUCLIDEAN_SPECIFIC projections (Fock conformal, Hopf bundle, stereographic, spectral action, Camporesi–Higuchi spinor lift) all require a full Krein-space lift to extend cleanly to signature $(3,1)$.

Sections 2–10 do not address signature $(3,1)$ (the Krein carrier of §11 does, at finite cutoff). They operate entirely on the Riemannian Camporesi–Higuchi triple at signature $(3,0)$ and record the modular-period identity inside that framework. The Sprint L0 audit identified an operator-system test of the Wick-rotation projection at signature $(3,0)$ — a modular Hamiltonian computed from a wedge-restricted Camporesi–Higuchi state — as the principal Sprint L1 falsifier ahead of the $(3,1)$ extension (itself beyond sprint scale). The present paper does not run that test: its generator is posited, so its period closure is automatic (withdrawn as a closure of the falsifier, 2026-09-28).

3.2 Footnote on the Nieuviarts twist morphism

A natural shortcut to the signature change would be the twisted-spectral-triple morphism of Nieuviarts [21–23], which constructs a pseudo-Riemannian spectral triple from an almost-commutative twisted Riemannian spectral triple, presented as an alternative to Wick rotation. The construction is mechanistically appealing but does *not* apply to the present S^3 case: Definition 2.2 of [22] requires a $\mathbb{Z}_2$ grading Γ that commutes with the algebra and anti-commutes with the Dirac, which exists only for *even* KO-dimension spectral triples. The S^3 Camporesi–Higuchi triple has KO-dimension 3 (mod 8) (odd); the restriction is structural rather than editorial (the author of [22] explicitly notes: "the fact that no such procedure for twisting odd-dimensional manifold's spectral triple has been found will be one of the justifications to focus on the study of even-dimensional manifolds"). The Lorentzian-extension path for the present construction therefore runs through the direct Bizi–Brouder–Besnard 2018 Krein-space lift, not through a twist morphism; this is the named Sprint L2 follow-up of the present author's framework documentation. A dedicated NO-GO analysis of the Nieuviarts route on $S^3 = \mathrm{SU}(2)$ is recorded in `debug/nieuviarts_scoping_memo.md` of the present author's internal materials.

4 The Bisognano–Wichmann modular structure at finite cutoff

We set up the operator-system-level analog of the Bisognano–Wichmann construction on the truncated Camporesi–Higuchi spectral triple. Three ingredients are required: (i) the hemispheric wedge projector P_W, (ii) the wedge-restricted Gibbs state ω_W at inverse temperature β, (iii) the local Hamiltonian H_{local} generating the wedge-modular evolution.

4.1 The hemispheric wedge projector

Definition 4.1 (Hemispheric wedge on $\mathcal{H}_{n_{\max}}$)**.** The *Hopf-axis equatorial reflection* on $\mathcal{H}_{n_{\max}}$ is the involution

$$R_{\text{polar}} : \psi_{n,l,m_j,\chi} \longmapsto \psi_{n,l,-m_j,\chi}, \tag{8}$$

acting as $m_j \mapsto -m_j$ on the spinor basis. The *hemispheric wedge projector* is

$$P_W := \tfrac{1}{2}(I + R_{\text{polar}}) \in B(\mathcal{H}_{n_{\max}}). \tag{9}$$

Proposition 4.2 (Wedge projector properties)**.** *The hemispheric wedge projector P_W satisfies:*

(a) $P_W^2 = P_W$ *(idempotent);*

(b) $P_W^* = P_W$ *(Hermitian);*

(c) $\mathrm{Tr}(P_W) = \dim \mathcal{H}_{n_{\max}}/2$ *(half-dimensional projection).*

Equivalently, $\mathcal{H}_{n_{\max}} = \mathcal{H}_W \oplus \mathcal{H}_{W'}$ *where* $\mathcal{H}_W = P_W \mathcal{H}_{n_{\max}}$ *has dimension* $\dim_W = \dim \mathcal{H}_{n_{\max}}/2$*, i.e.,* $\dim_W = 8, 20, 40, 70$ *at* $n_{\max} = 2, 3, 4, 5$*.*

Proof. (a) and (b) follow from $R_{\text{polar}}^2 = I$ and $R_{\text{polar}}^* = R_{\text{polar}}$, the latter holding because R_{polar} is a permutation matrix. For (c): on the full-Dirac spinor basis the values two_m_j are *odd integers*, so the involution $m_j \mapsto -m_j$ has no fixed points, i.e., R_{polar} has spectrum $\{+1, -1\}$ each with multiplicity $\dim \mathcal{H}_{n_{\max}}/2$. □

The hemispheric wedge $\mathcal{H}_W$ is the operator-system analog of the half-S^3 algebra $C^\infty(H_+)$ where $H_+ = \{\omega \in S^3 : \omega \cdot \hat{n} > 0\}$ is the open hemisphere of S^3 aligned with a chosen polar axis $\hat{n}$. The choice of axis is fixed canonically by alignment with the Hopf-base direction (see [35, §V.3] for the role of the Hopf-base axis in the present author's framework documentation, and Section 6 below for the role of the integer spectrum of two_m_j on this axis).

Remark 4.3 (On the choice of wedge)**.** Three candidates for the wedge sub-system were named in the architectural scoping (`debug/l1_modular_hamiltonian_architecture_memo.md` §3): (W1) the hemispheric wedge of Definition 4.1; (W2) the κ-preserving block of Paper 29's Dirac graph Rule A; (W3) the pendant-vertex-free subgraph of the scalar Fock graph. Candidate W1 is the cleanest Riemannian-side analog of the Wightman Rindler wedge (a region bounded by a horizon of codimension 1 in the underlying manifold); the polar rotation generator does *not* preserve it (it anticommutes with R_{polar}). Candidates W2 and W3 are graph-topological projections rather than spatial-region restrictions, and their modular structure either degenerates (W2) or has no clear "localised in a half-region" interpretation (W3). The present paper proceeds with W1 as the canonical choice. The dependence of the closure on the wedge choice is part of the open scope (Section 10).

4.2 The wedge-restricted Gibbs state

For a self-adjoint operator H on $\mathcal{H}_W$ and a positive inverse temperature $\beta > 0$, the Gibbs state $\omega_{\beta,H} : B(\mathcal{H}_W) \to \mathbb{C}$ is

$$\omega_{\beta,H}(O) = \frac{\mathrm{Tr}_{\mathcal{H}_W}\left(e^{-\beta H} O\right)}{\mathrm{Tr}_{\mathcal{H}_W}\left(e^{-\beta H}\right)}, \qquad O \in B(\mathcal{H}_W). \tag{10}$$

Restricted to the wedge sub-operator-system $\mathcal{O}_{W,n_{\max}} := P_W \mathcal{O}_{n_{\max}} P_W$, this is the natural Hilbert-space-internal analog of the Wightman Rindler-wedge KMS state.

The choice of H is the load-bearing structural input, and it is a *posit*: nothing below derives it from the Camporesi–Higuchi triple. We make the choice suggested by the form of the Wightman BW vacuum (Section 6 below):

$$H_{\text{local}} := K_\alpha^W / \beta, \tag{11}$$

where K_α^W is the geometric BW-α generator of Definition 5.1 below. With this choice, $\beta H_{\text{local}} = K_\alpha^W$ is itself β-independent at the algebra-action level, so the wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$ is the same for every β. This is a deliberate construction-level choice; an alternative choice $H_{\text{local}} = D_W$ (the wedge-restricted Camporesi–Higuchi Dirac) would lift the $\beta = 2\pi$ period to the spinor double-cover $-I$ rather than the identity operator $+I$ of Section 5 (the conjugation flow still closes, but at the double-cover sign). We return to this point in Section 7.

4.3 Four-witness unification and unit normalisation

The four physical witnesses (Hartle–Hawking, Sewell, Bisognano–Wichmann, Unruh) share the structural identity

$$\beta_W = 2\pi / \kappa_g(W), \tag{12}$$

where $\kappa_g(W)$ is the surface gravity of the bifurcate Killing horizon bounding the causal region W: $\kappa_g^{\text{HH}} = 1/(4M)$ for the Schwarzschild exterior of mass M, giving $\beta^{\text{HH}} = 8\pi M$; $\kappa_g^{\text{Sew}} = \kappa_g^{\text{HH}}$ for the abstract KMS-modular reading of the Hartle–Hawking state; $\kappa_g^{\text{BW}} = 1$ in natural rapidity units for the Rindler wedge, giving $\beta^{\text{BW}} = 2\pi$; $\kappa_g^{\text{Unruh}} = a$ for the Rindler wedge in proper-acceleration units, giving $\beta^{\text{Unruh}} = 2\pi/a$.

Unit normalisation. We adopt the BW rapidity-units convention $\kappa_g = 1$ (denoted *U-1* in the architectural scoping) as the BW witness. The four physical witnesses are then instantiated as parameterised BW with witness-specific κ_g values: BW $\to \kappa_g = 1$, HH $\to \kappa_g = 1/(4M)$, Sew $\to \kappa_g = 1/(4M)$, Unruh $\to \kappa_g = a$. At the operator level every instantiation uses the *same* generator K_α^W; κ_g enters only as a label (through $\beta = 2\pi/\kappa_g$ and $H_{\text{local}} = K_\alpha^W/\beta$), and cancels in βH_{local}. The period closure $\sigma_{2\pi}(O) = O$ is therefore κ_g-independent by construction (Sections 5–8), and the agreement of the four witnesses at finite cutoff is a consistency check of the bookkeeping, not a finite-cutoff theorem about them.

5 The geometric construction (BW-α)

The first construction on $\mathcal{T}_{n_{\max}}$ is the geometric BW-α generator, obtained by *positing* that the role of the boost is played by a relabelling built from the spatial rotation generator about the polar axis (which itself does not preserve the wedge, Definition 5.1). On the Riemannian S^3 there is no boost; the construction is a stand-in, not an identification.

5.1 The geometric generator

Definition 5.1 (BW-α geometric generator on the wedge)**.** The *BW-α geometric generator* on the wedge sub-Hilbert space $\mathcal{H}_W = P_W \mathcal{H}_{n_{\max}}$ is the operator

$$K_\alpha^W := \text{diag}(\text{two_}m_j) \quad \text{on} \quad \mathcal{H}_W, \tag{13}$$

acting on each wedge basis state $\psi_{n,l,|m_j|,\chi}$ (with m_j chosen positive on the wedge P_W) by multiplication by two$_m_j$, the doubled m-quantum-number. It is a relabelling, not the compression $P_W(2J_z)P_W$: $2J_z$ anticommutes with R_{polar}, so that compression is zero.

Remark 5.2 (Why two_m_j rather than m_j)**.** On the full-Dirac spinor basis $\{\psi_{n,l,m_j,\chi}\}$, the quantum number m_j takes half-integer values $\{-l-1/2, -l+1/2, \ldots, l+1/2\}$. To obtain an integer spectrum — the load-bearing property for the bit-exact period closure of Theorem 5.4 — we work with the doubled generator two_$m_j = 2m_j$, taking odd integer values. This is the standard half-integer-spin doubling convention from representation theory of SU(2). The factor of 2 does not affect the modular flow at the period (modular flow is invariant under overall rescaling of K by integer factors), but it makes the integer-spectrum property manifest.

Proposition 5.3 (Integer spectrum)**.** *The spectrum of K_α^W on $\mathcal{H}_W$ is contained in the odd integers $\{-(2l+1), -(2l-1), \ldots, +(2l-1), +(2l+1)\}$ over $l \in \{0, 1, \ldots, n_{\max}-1\}$. In particular,* $\mathrm{Spec}(K_\alpha^W) \subset \mathbb{Z}$.

Proof. On the wedge P_W, m_j takes values $\{1/2, 3/2, \ldots, l+1/2\}$ (positive half-integers up to $l+1/2$). The doubled generator two_m_j takes values $\{1, 3, \ldots, 2l+1\}$, all odd positive integers. Including the wedge-anti-wedge structure via the full-Dirac basis would give the full range $\{-(2l+1), \ldots, +(2l+1)\}$ in odd integer steps; on the wedge itself we have the positive sub-range. □

5.2 The geometric modular flow

The geometric BW-α flow (a modular flow only if O0 identifies K_α^W as a modular Hamiltonian) on the wedge sub-operator-system $\mathcal{O}_{W,n_{\max}}$ is the inner-derivation flow

$$\sigma_t^\alpha(O) \;:=\; e^{itK_\alpha^W}\, O\, e^{-itK_\alpha^W}, \qquad O \in \mathcal{O}_{W,n_{\max}},\ t \in \mathbb{R}. \tag{14}$$

This is well-defined as a strongly continuous one-parameter group of unital $*$-automorphisms on $B(\mathcal{H}_W)$. Its restriction to $\mathcal{O}_{W,n_{\max}}$ may leave the operator system (since $\mathcal{O}_{W,n_{\max}}$ is not multiplicatively closed); we return to this point in Section 5.3.

Theorem 5.4 (BW-α period closure at finite cutoff, [SYMBOLIC])**.** *For every $n_{\max} \geq 1$ and every $O \in B(\mathcal{H}_W)$, the BW-α modular flow (14) satisfies the period closure*

$$\sigma_{2\pi}^\alpha(O) \;=\; O \tag{15}$$

identically, i.e., as a bit-exact operator identity.

Proof. By Proposition 5.3, K_α^W is diagonal with integer eigenvalues $\{n_1, n_2, \ldots, n_{\dim_W}\}$ in its eigenbasis. The unitary $e^{i\cdot 2\pi\cdot K_\alpha^W}$ is then diagonal with eigenvalues $e^{i\cdot 2\pi\cdot n_j} = 1$ for every j (since each n_j is an integer): $e^{i\cdot 2\pi\cdot K_\alpha^W} = I_{\dim_W}$ identically. Hence $\sigma_{2\pi}^\alpha(O) = I \cdot O \cdot I = O$ for every O. □

Remark 5.5 (What Theorem 5.4 does not test)**.** The proof uses only that K_α^W is Hermitian with integer spectrum; it uses nothing about the wedge, the Camporesi–Higuchi Dirac, or any state. The same closure holds for *every* integer-spectrum generator and *every* operator O, so it cannot distinguish a Bisognano–Wichmann modular flow from any other 2π-periodic unitary group. (Concretely: replacing K_α^W in the supporting module by an unrelated integer-spectrum matrix leaves green all eighteen period-closure and cross-witness tests of `tests/test_modular_hamiltonian.py` exercised in that fire test, 2026-09-28.) The theorem is a correct elementary fact; it is not evidence that K_α^W is the modular Hamiltonian of a physically distinguished wedge state.

Numerical verification. Theorem 5.4 is an exact algebraic identity at the level of the symbolic construction. In a floating-point computation, the matrix exponential $e^{i\cdot 2\pi\cdot K_\alpha^W}$ is computed by diagonalisation and then exponentiation of the eigenvalues, so the identity holds to within the floating-point precision of the diagonalisation step. We have computed the maximum periodicity residual

$$r_W^\alpha(O) \;:=\; \|\sigma_{2\pi}^\alpha(O) - O\|_F \tag{16}$$

for the first five non-identity multipliers $O \in \mathcal{O}_{n_{\max}}$ (restricted to the wedge), at $n_{\max} \in \{2,3,4,5\}$. The results are recorded in Table 1.

Table 1: Maximum BW-α periodicity residual $\max_O r_W^\alpha(O)$ at finite cutoff. Computed in double-precision floating point on the full-Dirac basis; residuals scale as $O(\sqrt{\dim \mathcal{H}_{n_{\max}}}\cdot\varepsilon_{\text{machine}})$, i.e., pure round-off accumulation with no structural drift. The verdict column reproduces the supporting module's label; it records that the automatic closure was reproduced, not an identification (Remark 5.5). Source data: `debug/data/l1_modular_hamiltonian_results.json`.

$n_{\max}$	$\dim \mathcal{H}_{n_{\max}}$	$\dim_W$	$\max_O r_W^\alpha(O)$	verdict
2	16	8	1.80×10^{-16}	STRONG_IDENTIFICATION
3	40	20	4.76×10^{-16}	STRONG_IDENTIFICATION
4	80	40	9.33×10^{-16}	STRONG_IDENTIFICATION
5	140	70	1.59×10^{-15}	STRONG_IDENTIFICATION

The $n_{\max} = 3$ row is the one at which $\dim \mathcal{H}_3 = 40$ numerically coincides with $g_3^{\text{Dirac}} = \Delta^{-1} = 40$ (different objects that agree only at $n_{\max} = 3$, as noted above), the selection-principle cutoff of the present author's Paper 2 [33] fine-structure-constant observation. Nothing about the closure is special at this cutoff: it holds at every $n_{\max}$ for the integer-spectrum reason above.

5.3 Operator-system non-leakage

A separate concern flagged in the architectural scoping (`debug/l1_modular_hamiltonian_architecture_memo.` §10a) is that the truncated operator system $\mathcal{O}_{n_{\max}}$ is $*$-closed but not multiplicatively closed: the modular flow σ_t^α acts on all of $B(\mathcal{H}_W)$, and its restriction to $\mathcal{O}_{W,n_{\max}}$ might leak out. The integer-spectrum closure Theorem 5.4 resolves this without further work: $\sigma_{2\pi}^\alpha(O) = O$ for every O, so the flow at the period $t = 2\pi$ leaves every element of $\mathcal{O}_{W,n_{\max}}$ inside $\mathcal{O}_{W,n_{\max}}$ trivially. At intermediate $t \in (0, 2\pi)$ the flow may leave the operator system, but the period closure (the modular identity we test) is unaffected.

6 The Tomita–Takesaki construction (BW-γ)

The second construction on $\mathcal{T}_{n_{\max}}$ is the Tomita–Takesaki BW-γ reading, obtained by polar decomposition of the modular S operator on the GNS Hilbert–Schmidt space of the wedge-restricted Gibbs state. Because that state is built from the posited generator, $\rho_W = e^{-K_\alpha^W}/Z$, this construction returns K_α^W (Tomita's theorem for a Gibbs state) and is *not* an independent check on it; its value here is that it exhibits the GNS/Tomita machinery explicitly at finite cutoff, ready for the [OPEN] case in which the state is not built from K.

6.1 The GNS Hilbert–Schmidt space

Let $\rho := e^{-K_\alpha^W}/Z$ be the wedge density matrix (Section 4.2 with $H_{\text{local}} = K_\alpha^W/\beta$, so $\beta H_{\text{local}} = K_\alpha^W$ is β-independent). For the von Neumann algebra $\mathcal{A}_W := B(\mathcal{H}_W) \cong M_{\dim_W}(\mathbb{C})$ (Type $\mathrm{I}_{\dim_W}$ factor) and the faithful normal state $\omega(O) := \mathrm{Tr}_{\mathcal{H}_W}(\rho\, O)$, the GNS triple $(\pi_\omega, \mathcal{H}_{\text{GNS}}, \Omega)$

is realised on the Hilbert–Schmidt space

$$\mathcal{H}_{\mathrm{GNS}} \;=\; M_{\dim_W}(\mathbb{C}) \;\cong\; \mathbb{C}^{\dim_W^2}, \qquad \langle X, Y\rangle_{\mathrm{GNS}} \;:=\; \mathrm{Tr}(X^* Y), \tag{17}$$

with cyclic vector $\Omega := \rho^{1/2}$ (the canonical purification of ρ). The left action of $a \in \mathcal{A}_W$ is $L_a(X) = aX$, the right action of a from the commutant $\mathcal{A}_W^{\mathrm{op}}$ is $R_a(X) = Xa$. Numerically, $\dim \mathcal{H}_{\mathrm{GNS}} = \dim_W^2 = 64, 400, 1600, 4900$ at $n_{\max} = 2, 3, 4, 5$.

6.2 The modular S operator and polar decomposition

The Tomita modular operator $S : \pi_\omega(a)\Omega \mapsto \pi_\omega(a^*)\Omega$ is the antilinear closure of the conjugate-linear involution $a\Omega \mapsto a^*\Omega$ on the cyclic-separating orbit. In the Hilbert–Schmidt realisation (17), the cyclic orbit is the set of $X = a\rho^{1/2}$ for $a \in \mathcal{A}_W$ (equivalently, all of $\mathcal{H}_{\mathrm{GNS}}$ if ρ is positive definite, which is the present case). The operation $a\Omega \mapsto a^*\Omega$ is then

$$S(X) \;=\; \rho^{1/2}\,(X\rho^{-1/2})^* \;=\; \rho^{1/2}\,\rho^{-1/2}\,X^* \;=\; X^*, \tag{18}$$

where the second equality uses $\rho^{1/2}\rho^{-1/2} = I$ and the fact that ρ is self-adjoint, so the $\rho^{1/2}$ on the left and $\rho^{-1/2}$ on the right cancel against X^* on a tracial cyclic vector. The polar decomposition $S = J_{\mathrm{TT}} \cdot \Delta^{1/2}$ gives:

$$\Delta \;=\; S^* S, \qquad \Delta(X) \;=\; \rho\, X\, \rho^{-1} \tag{19}$$

acting on $X \in \mathcal{H}_{\mathrm{GNS}}$, and

$$J_{\mathrm{TT}}(X) \;=\; \rho^{1/2}\, X^*\, \rho^{-1/2} \tag{20}$$

acting antilinearly on $\mathcal{H}_{\mathrm{GNS}}$. In the column-stacked matrix form $\mathrm{vec}(X) \in \mathbb{C}^{\dim_W^2}$, the modular operator Δ is

$$\Delta \;=\; (\rho^{-1})^T \otimes \rho \quad \text{acting on} \quad \mathrm{vec}(X) \in \mathbb{C}^{\dim_W^2}. \tag{21}$$

The *modular Hamiltonian* is then

$$K_{\mathrm{TT}} \;:=\; -\log \Delta, \tag{22}$$

and the *modular flow on the algebra* is

$$\sigma_t^{\mathrm{TT}}(a) \;:=\; \Delta^{it}(a) \;=\; \rho^{it}\, a\, \rho^{-it} \;=\; e^{-it\,K_\alpha^W}\, a\, e^{+it\,K_\alpha^W}, \tag{23}$$

where the last equality uses $\rho = e^{-K_\alpha^W}/Z$ and the Z factor cancels in the conjugation.

6.3 The Tomita antilinear conjugation

A structurally important caution is that the Tomita modular conjugation J_{TT} defined in (20) is *categorically different* from the Connes KO-dim 3 charge conjugation J_{GV} on the underlying Camporesi–Higuchi spectral triple.

Proposition 6.1 ($J_{\mathrm{TT}} \neq J_{\mathrm{GV}}$)**.** *The Tomita conjugation J_{TT} on $\mathcal{H}_{\mathrm{GNS}}$ and the Connes real structure J_{GV} on $\mathcal{H}_{n_{\max}}$ are categorically distinct antilinear operators with the following signature distinction:*

$$J_{\mathrm{TT}}^2 \;=\; +I, \qquad J_{\mathrm{GV}}^2 \;=\; -I. \tag{24}$$

Proof. For $J_{\mathrm{TT}}^2 = +I$: applying (20) twice gives

$$J_{\mathrm{TT}}^2(X) \;=\; J_{\mathrm{TT}}\big(\rho^{1/2}\, X^*\, \rho^{-1/2}\big) \;=\; \rho^{1/2}\,(\rho^{1/2}\, X^*\, \rho^{-1/2})^*\, \rho^{-1/2} \;=\; \rho^{1/2}\,\rho^{-1/2}\, X\, \rho^{1/2}\,\rho^{-1/2} \;=\; X,$$

where the second-to-last step uses $(\rho^{1/2})^* = \rho^{1/2}$ and $(\rho^{-1/2})^* = \rho^{-1/2}$ (both are self-adjoint), and the last step uses $\rho^{1/2}\rho^{-1/2} = I$ twice. For $J_{\mathrm{GV}}^2 = -I$: the Connes real structure on a KO-dim 3 spectral triple satisfies $J^2 = -I$ by convention (cf. Paper 32 § IV `rem:axiom_audit`). □

The two antilinear conjugations coexist on the same Hilbert spaces: J_{GV} on $\mathcal{H}_{n_{\max}}$, J_{TT} on $\mathcal{H}_{\mathrm{GNS}} \cong B(\mathcal{H}_W)$. Conflating them (e.g., by importing `geovac.real_structure.RealStructure` into the modular-Hamiltonian construction) would silently produce algebraic errors, since $J_{\mathrm{TT}}^2 - (-I) = 2I \neq 0$.

Remark 6.2 (State-dependence vs. kinematic intrinsicness)**.** J_{GV} is intrinsic to the spectral triple at construction time: once $(\mathcal{A}, \mathcal{H}, D, J, \gamma)$ is specified, J_{GV} is fixed by the KO-dim convention. J_{TT} is *state-dependent*: each KMS state ω defines its own $J_{\mathrm{TT}}^{(\omega)}$ via Tomita's theorem on the GNS triple of ω. The state-dependence is consistent with the standard Tomita–Takesaki theorem [26, 27].

6.4 Bit-exact period closure at the Tomita level

Theorem 6.3 (BW-γ period closure at finite cutoff, [SYMBOLIC]; automatic, as Theorem 5.4)**.** *For every $n_{\max} \geq 1$ and every $a \in B(\mathcal{H}_W)$, the Tomita–Takesaki modular flow (23) satisfies the period closure*

$$\sigma_{2\pi}^{\mathrm{TT}}(a) \;=\; a \tag{25}$$

identically, with maximum periodicity residual

$$r_W^{\mathrm{TT}}(O) \;:=\; \|\sigma_{2\pi}^{\mathrm{TT}}(O) - O\|_F$$

scaling as $O(\sqrt{\dim_W} \cdot \varepsilon_{\mathrm{machine}})$ in floating-point computation.

Proof. From (23), $\sigma_{2\pi}^{\mathrm{TT}}(a) = e^{-i\cdot 2\pi\cdot K_\alpha^W}\, a\, e^{+i\cdot 2\pi\cdot K_\alpha^W}$. By Proposition 5.3, K_α^W has integer spectrum, so $e^{\pm i\cdot 2\pi\cdot K_\alpha^W} = I$ identically. Hence $\sigma_{2\pi}^{\mathrm{TT}}(a) = a$. □

Numerical verification. In floating-point computation, the modular operator Δ is built on the GNS Hilbert–Schmidt space $\mathcal{H}_{\mathrm{GNS}} \cong \mathbb{C}^{\dim_W^2}$ via the tensor-product form (21), with each $\rho^{\pm 1}$ assembled by matrix exponential of $\pm K_\alpha^W$ and a numerical division by Z. The modular flow σ_t^{TT} is then computed by acting Δ^{it} on each input operator a written in vec form, then reshaping. We have computed the maximum periodicity residual $r_W^{\mathrm{TT}}(O)$ at $n_{\max} \in \{2, 3, 4, 5\}$; results are in Table 2.

Table 2: Maximum BW-γ Tomita-level periodicity residual $\max_O r_W^{\mathrm{TT}}(O)$ at finite cutoff, computed via polar decomposition on the GNS Hilbert–Schmidt space. The slightly larger residual at $n_{\max} = 5$ (vs. Table 1) is pure numerical conditioning from the $\dim_W^2 \times \dim_W^2 = 4900 \times 4900$ matrix diagonalisation of Δ with extreme eigenvalue ratios λ_i/λ_j; no structural drift. *Module label, reproduced as computed; it records the automatic closure, not an identification (the state is built from the posited generator). Source data: `debug/data/l1_tighten_tomita_results.json`.

$n_{\max}$	$\dim_W$	$\dim \mathcal{H}_{\mathrm{GNS}}$	$\max_O r_W^{\mathrm{TT}}(O)$	verdict
2	8	64	1.09×10^{-16}	UNIFIED_STRONG*
3	20	400	1.07×10^{-15}	UNIFIED_STRONG*
4	40	1600	3.03×10^{-15}	UNIFIED_STRONG*
5	70	4900	4.79×10^{-15}	UNIFIED_STRONG*

The $J_{\mathrm{TT}}^2 = +I$ identity of Proposition 6.1 is verified numerically at every $n_{\max}$ to machine precision (residuals $\leq 7 \times 10^{-17}$): this distinguishes J_{TT} from J_{GV} at the genuine state-dependent level, not just at the canonical $U = I$ sanity case.

6.5 Spectrum of K_{TT}

Proposition 6.4 (Spectrum of K_{TT} on $\mathcal{H}_{\mathrm{GNS}}$)**.** *The eigenvalues of $K_{\mathrm{TT}} = -\log\Delta$ on $\mathcal{H}_{\mathrm{GNS}}$ are $\{n_j - n_k : 1 \le j,k \le \dim_W\}$, where $\{n_j\}$ is the integer spectrum of K_α^W. In particular:*

(a) $\mathrm{Spec}(K_{\mathrm{TT}}) \subset \mathbb{Z}$ *(integer spectrum on $\mathcal{H}_{\mathrm{GNS}}$);*

(b) K_{TT} *has at least* $\dim_W$ *zero eigenvalues (from the diagonal $j=k$ pairs);*

(c) The maximum eigenvalue of K_{TT} *is* $2\max_j n_j = 2(2(n_{\max}-1)+1) = 2(2n_{\max}-1)$*; for* $n_{\max} = 2$ *this is* ≤ 4*, and the maximum is exactly* 2 *on the wedge (since* $\max_W \mathrm{two_}m_j = +1$ *at* $n_{\max} = 2$*).*

Proof. By (21), $\Delta = (\rho^{-1})^T \otimes \rho$ in the vec representation, with $\rho = e^{-K_\alpha^W}/Z$. The eigenvalues of ρ are $\{e^{-n_j}/Z\}_{j=1}^{\dim_W}$, so the eigenvalues of Δ are $\{e^{n_k}\cdot e^{-n_j}\}_{j,k} = \{e^{n_k-n_j}\}_{j,k}$. Hence $K_{\mathrm{TT}} = -\log\Delta$ has eigenvalues $\{n_j - n_k\}$, which are integer differences of the integer spectrum of K_α^W. The diagonal $j=k$ pairs contribute $n_j - n_j = 0$, giving at least $\dim_W$ zero eigenvalues (more if K_α^W has spectral degeneracies). □

The integer-spectrum property of Proposition 6.4(a) is the structural reason the bit-exact closure of Theorem 6.3 holds: $e^{i\cdot 2\pi\cdot K_{\mathrm{TT}}} = I$ on $\mathcal{H}_{\mathrm{GNS}}$ by $e^{i\cdot 2\pi\cdot m} = 1$ for integer m, so the modular flow Δ^{it} acting at $t = 2\pi$ collapses to the identity, and $\sigma_{2\pi}^{\mathrm{TT}}(a) = I\cdot a\cdot I = a$ for every a. This is the same mechanism as Theorem 5.4 but realised at the GNS-Hilbert–Schmidt level.

6.6 State-space GH bound cross-check

The closure of Theorem 6.3 is a finite-cutoff identity, not a propinquity limit statement. To confirm consistency with Paper 38's qualitative-rate state-space-GH bound (7), we compute the ratio

$$\frac{\max_O r_W^{\mathrm{TT}}(O)}{\gamma_{n_{\max}}^{(\mathrm{L2})}}$$

where $\gamma_{n_{\max}}^{(\mathrm{L2})}$ is the L2 mass-concentration moment of the central spectral Fejér kernel on SU(2) ([40, Lemma L2]): $\gamma_2^{(\mathrm{L2})} = 2.0746$, $\gamma_3^{(\mathrm{L2})} = 1.6101$, $\gamma_4^{(\mathrm{L2})} = 1.3223$, $\gamma_5^{(\mathrm{L2})} = 1.1302$. The ratios are 8.7×10^{-17}, 3.0×10^{-16}, 7.1×10^{-16}, 1.4×10^{-15}: several orders of magnitude below the qualitative-rate bound at every tested $n_{\max}$. The comparison is a sanity check only and carries no content: the closure is exact for the integer-spectrum reason, so its residual is pure round-off and would be equally small for any integer-spectrum generator, whatever the metric convergence rate.

7 Unified closure

The two constructions of Sections 5 and 6 are not the same operator on $\mathcal{H}_W$ versus $\mathcal{H}_{\mathrm{GNS}}$ (the GNS construction lives on a larger Hilbert space). The modular flows they generate on the algebra $B(\mathcal{H}_W)$ are the same flow up to time reversal, because the state was built from K_α^W; they are not structurally independent realisations. This section makes the relation precise.

7.1 Flow conjugacy

Theorem 7.1 (BW-α / BW-γ flow conjugacy, [SYMBOLIC]; Tomita's theorem for a Gibbs state)**.** *For every $a \in B(\mathcal{H}_W)$ and every $t \in \mathbb{R}$,*

$$\sigma_t^{\mathrm{TT}}(a) \;=\; \sigma_{-t}^{\alpha}(a), \tag{26}$$

where σ_t^α is the geometric BW-α flow (14) and σ_t^{TT} is the Tomita–Takesaki BW-γ flow (23).

Proof. By (23), $\sigma_t^{\mathrm{TT}}(a) = e^{-itK_\alpha^W}\, a\, e^{+itK_\alpha^W}$. By (14), $\sigma_{-t}^{\alpha}(a) = e^{-itK_\alpha^W}\, a\, e^{+itK_\alpha^W}$. The two expressions are literally identical. □

Consequence at the period. At $t = 2\pi$, equation (26) reads $\sigma_{2\pi}^{\mathrm{TT}}(a) = \sigma_{-2\pi}^{\alpha}(a) = \sigma_{2\pi}^{\alpha}(a)^{-1}$, where the last equality uses that the flow σ_t^α is a one-parameter group with the inverse at $t = 2\pi$ being the flow at $t = -2\pi$. But by Theorem 5.4, $\sigma_{2\pi}^{\alpha}(a) = a = a^{-1}$ trivially (this is an operator equation, not a group inverse), so $\sigma_{2\pi}^{\mathrm{TT}}(a) = a$ as well, consistent with Theorem 6.3. The two closures are the *same* closure under the conjugacy (26), which is why their agreement carries no independent evidential weight.

Numerical confirmation at general t. At a general non-period t (e.g., $t = 1$), the two flows σ_t^{TT} and σ_t^α are distinct (the conjugacy (26) relates σ_t^{TT} to σ_{-t}^α, not to σ_t^α). We have verified this numerically: the average cross-flow difference $\langle \|\sigma_1^{\mathrm{TT}}(O) - \sigma_1^\alpha(O)\|_F \rangle_O$ over the same five test operators is ≈ 0.16 at $n_{\max} = 2$, clearly non-zero. The sign-flipped comparison $\langle \|\sigma_1^{\mathrm{TT}}(O) - \sigma_{-1}^\alpha(O)\|_F \rangle_O$ gives $\approx 4 \times 10^{-17}$, machine precision. The conjugacy (26) is therefore verified bit-exactly at general t, not just at the period.

7.2 The posited local Hamiltonian

The unified closure rests on the deliberate choice $H_{\mathrm{local}} = K_\alpha^W/\beta$ (equation (11)) for the local Hamiltonian generating the wedge KMS state. This is a construction-level choice, not a derivation: we wrote down the wedge KMS state as $\rho_W = e^{-K_\alpha^W}/Z$ rather than (say) $\rho_W = e^{-\beta D_W}/Z$ for the wedge-restricted Camporesi–Higuchi Dirac. The motivation is structural (we want the modular flow to close with period 2π, as the Bisognano–Wichmann flow does), but it is *a* choice of local Hamiltonian, not the unique choice. The generator K_α^W (the relabelling of Definition 5.1) is *posited*, and $\rho_W = e^{-K_\alpha^W}/Z$ is its unit-β Gibbs state, not the Bisognano–Wichmann-normalised state, which (KMS parameter 2π in the rotation angle) would be $e^{-\pi K_\alpha^W}/Z$, the same relabelling applied to $e^{-2\pi J_z}$ (Paper 50, remark on the normalisation of K_α and ρ_W).

Two remarks place the choice; neither derives it:

(I) Formal resemblance to the Wightman BW vacuum. In the continuum Bisognano–Wichmann theorem, the local Hamiltonian on the Rindler wedge is the boost generator K_{boost}, and the Wightman vacuum restricted to the wedge algebra is the Gibbs state $\rho_W = e^{-2\pi K_{\mathrm{boost}}}/Z$ at inverse temperature $\beta = 2\pi$. Our choice $H_{\mathrm{local}} = K_\alpha^W/\beta$ copies this *form* (not its normalisation, see above): $\beta H_{\mathrm{local}} = K_\alpha^W$, with the spatial rotation generator around the Hopf-base axis standing in for the boost, and $\rho_W = e^{-K_\alpha^W}/Z$. In the continuum the Gibbs form is a *theorem* about the vacuum; here it is the definition of the state, so the resemblance is by design and is not evidence that ρ_W is the restriction of any distinguished global state.

(II) The intrinsic Dirac lifts the period to $-I$. If we had chosen $H_{\mathrm{local}} = D_W$ (the wedge-restricted Camporesi–Higuchi Dirac), whose eigenvalues are $\chi \cdot (n + 1/2)$ (half-integer on the spinor basis), then the *operator-level* period lift $e^{i\cdot 2\pi \cdot D_W} = -I$ — the global -1 of the spinor double cover — whereas the posited K_α^W has odd-integer spectrum and $e^{i\cdot 2\pi \cdot K_\alpha^W} = +I$, the identity operator. We emphasise the precise sense of the distinction: the modular flow $\sigma_t(\cdot) = U_t(\cdot)U_t^{-1}$ acts by *conjugation*, in which a global scalar phase cancels, so the flow-level period closure $\sigma_{2\pi}(O) = O$ still holds with D_W (the $-I$ cancels: $(-I)\,O\,(-I) = O$; we verified this bit-exactly, residual $< 10^{-13}$ at $n_{\max} = 2, 3, 4$; `tests/test_modular_hamiltonian.py`). The distinction is therefore *not* at the flow level but at the operator level: only K_α^W lifts the $\beta = 2\pi$ period to the identity operator $+I$; D_W lifts it to the double-cover $-I$.

This is an elementary observation (integer versus half-integer spectrum), not a selection theorem. The framework's intrinsic Dirac D_{CH} is the canonical spectral-action object on the

Camporesi–Higuchi triple, and one might expect it to play the role of the local Hamiltonian for a natural KMS state on the truncated operator system. The above shows only that, while either choice generates a 2π-periodic conjugation flow, the posited generator K_α^W lifts the period to $+I$ and the wedge-restricted Dirac D_W to the spinor double-cover sign $-I$; which (if either) is the modular generator of a Camporesi–Higuchi wedge state is the [OPEN] question O0. The two play complementary roles: D_{CH} is the spectral-action object (the Connes–Chamseddine input for the heat-kernel expansion); K_α^W is the wedge-modular-Hamiltonian object (the Tomita–Takesaki input for the modular-flow construction). On the round S^3 they happen to agree up to an overall scale (both are diagonal in the spinor basis with simple spectral structure), but only one of them has the integer spectrum that lifts the $\beta = 2\pi$ period to the identity operator (the same integer-spectrum / $e^{2\pi i K} = +I$ structure that underlies the compact-boost reading of Section 10).

Remark 7.2 (A speculative reading)**.** We note, but do not pursue here, the possibility that the distinction between the spectral-action Dirac D_{CH} and the modular-Hamiltonian generator K_α^W is structurally significant for the framework's interpretation. In the spectral-action paradigm (Chamseddine–Connes [7]), the Dirac is the universal object from which all physics is read; in the thermal-time paradigm (Connes–Rovelli [9]), the modular Hamiltonian of an algebraic state is the physical-time generator. On the round S^3 Camporesi–Higuchi triple at the wedge KMS state $\rho_W = e^{-K_\alpha^W}/Z$, these are not the same operator. The structural status of this distinction — whether it is a peculiarity of the round S^3 truncation or a deeper feature of the framework's spectral content — is open. We flag it as one of the open questions in Section 10.

7.3 Cross-witness collapse at the algebra-action level

The witness agreement mentioned in Section 4.3 holds by construction at the algebra-action level. With $H_{\mathrm{local}} = K_\alpha^W/\beta$, the density matrix is

$$\rho_W \;=\; \frac{e^{-\beta H_{\mathrm{local}}}}{Z} \;=\; \frac{e^{-K_\alpha^W}}{Z} \tag{27}$$

independent of β. The Tomita modular operator Δ constructed from ρ_W via (21) and the modular Hamiltonian $K_{\mathrm{TT}} = -\log\Delta$ are therefore also β-independent. All four witness instantiations (BW, HH, Sew, Unruh) therefore produce the *same* modular operator Δ and the same K_{TT}: they are the same K_α^W with κ_g relabelled, and κ_g drops out by construction. This is not a finite-cutoff statement about the four witnesses.

We have verified this numerically at $n_{\max} \in \{2,3,4,5\}$ across the six witness instantiations (BW witness at $\kappa_g = 1$; Hartle–Hawking at $M = 1$, $\kappa_g = 1/4$; Hartle–Hawking at $M = 2$, $\kappa_g = 1/8$; Sewell at $M = 1$, $\kappa_g = 1/4$; Unruh at $a = 1$, $\kappa_g = 1$; Unruh at $a = 2$, $\kappa_g = 2$). The cross-witness consistency residual is exactly 0 at every tested $n_{\max}$ between any pair of witnesses, as it must be: the six runs compute the same matrix, so the check confirms the bookkeeping, not a physical coincidence.

8 Four witnesses, one construction (by construction)

The numerical content of the witness agreement is summarised in Table 3 and its (elementary) structural content is the following corollary.

Corollary 8.1 (Single operator-system construction for the four witnesses, [SYMBOLIC]; by construction)**.** *Under the unit normalisation U-1 and the wedge KMS state choice* $\rho_W = e^{-K_\alpha^W}/Z$*, the operator-system-level realisations of the Hartle–Hawking, Sewell, Bisognano–Wichmann, and Unruh witnesses on the truncated Camporesi–Higuchi triple* $\mathcal{T}_{n_{\max}}$ *collapse to a single construction. The modular operator* Δ *and the modular Hamiltonian* K_{TT} *are bit-identical across the six*

Table 3: Cross-witness consistency at the BW-α level (left columns) and at the BW-γ Tomita level (right columns). Cross-consistency residual is the difference between the period residual of each witness and the BW-witness period residual at the same $n_{\max}$: exactly 0 across all six witnesses at every tested $n_{\max}$. Source data: `debug/data/l1_modular_hamiltonian_results.json`, `debug/data/l1_tighten_tomita_results.json`.

$n_{\max}$	BW ($\kappa_g=1$)	$\mathrm{HH}_{M=1}$ ($\kappa_g=1/4$)	$\mathrm{HH}_{M=2}$ ($\kappa_g=1/8$)	$\mathrm{Sew}_{M=1}$ ($\kappa_g=1/4$)	$\mathrm{Unruh}_{a=1}$ ($\kappa_g=1$)	$\mathrm{Unruh}_{a=2}$ ($\kappa_g=2$)
BW-α residual (Table 1):						
2	1.80e-16	1.80e-16	1.80e-16	1.80e-16	1.80e-16	1.80e-16
3	4.76e-16	4.76e-16	4.76e-16	4.76e-16	4.76e-16	4.76e-16
4	9.33e-16	9.33e-16	9.33e-16	9.33e-16	9.33e-16	9.33e-16
5	1.59e-15	1.59e-15	1.59e-15	1.59e-15	1.59e-15	1.59e-15
BW-γ Tomita residual (Table 2):						
2	1.09e-16	1.09e-16	1.09e-16	1.09e-16	1.09e-16	1.09e-16
3	1.07e-15	1.07e-15	1.07e-15	1.07e-15	1.07e-15	1.07e-15
4	3.03e-15	3.03e-15	3.03e-15	3.03e-15	3.03e-15	3.03e-15
5	4.79e-15	4.79e-15	4.79e-15	4.79e-15	4.79e-15	4.79e-15

witness instantiations. The witness-specific physics enters only through the parameterisation of κ_g, which is invisible at the algebra-action level and at the period closure $\sigma_{2\pi}(O) = O$.

Proof. The density matrix $\rho_W = e^{-K_\alpha^W}/Z$ is β-independent (Section 7.3). The Tomita modular operator Δ on the GNS Hilbert–Schmidt space is constructed entirely from ρ_W via (21), and the modular Hamiltonian $K_{\mathrm{TT}} = -\log\Delta$ depends only on Δ. Both are therefore β-independent. β-independence at the algebra-action level immediately implies identical period closure at $t = 2\pi$ for all six witness instantiations. □

Scope of the corollary. The corollary is true because the six instantiations are the same construction with a relabelled parameter; it is not a finite-cutoff theorem about Hartle–Hawking, Sewell, Bisognano–Wichmann, or Unruh. The witness-specific physical content (mass of the black hole, proper acceleration of the Rindler observer) enters only through the label κ_g, which the construction never uses. The 2π of the period closure comes from the integer spectrum of the posited $2J_z$; it is *not* shown here to be the same 2π as the M1 volume factor of the Paper 38 Fejér-moment constant (7), and the closure is not the operator-system realisation of the M1 submechanism. The metric-functional-level reading ([37, Rem. rem:bisognano_wichmann_reading], Sprint TD Track 4, Sprint Unruh-pendant), which matches the framework's 2π on a temporal S^1_τ with the modular-flow / boost-orbit period of the four witnesses via the published Wick-rotation chain, is unaffected and is *not* lifted by the present paper: the dependence on the Wick-rotation prescription remains, pending the [OPEN] derivation of K from a wedge-restricted Camporesi–Higuchi state.

Remark 8.2 (Hilbert–Schmidt orthogonality of H_{local} and D_W)**.** For every witness w the posited local Hamiltonian $H^{(w)}_{\mathrm{local}} = K^W_\alpha/\beta^{(w)}$ and the wedge-restricted Camporesi–Higuchi Dirac D^{GV}_W are Hilbert–Schmidt orthogonal, $\mathrm{Tr}\big((H^{(w)}_{\mathrm{local}})^\dagger D^{\mathrm{GV}}_W\big) = 0$, so that $\|H^{(w)}_{\mathrm{local}} - D^{\mathrm{GV}}_W\|^2_F = (\kappa_g^{(w)})^2 S(n_{\max})/(4\pi^2) + D(n_{\max})$ with S, D polynomials in $n_{\max}$. This is a chirality-parity identity, proved in Theorem 11.14 and Corollary 11.15 (§11.7): $H^{(w)}_{\mathrm{local}}$ is diagonal and independent of the chirality χ, while D^{GV}_W is chirality-diagonal with χ-paired eigenvalues of opposite sign, so

the pairs cancel in the trace. The identity is β-independent, the $1/\pi^2$ of the residual being the square of the posited normalisation $\beta = 2\pi/\kappa_g$. It does not "underlie" the six-witness agreement of Corollary 8.1, which holds by construction, and does not bear on the period closure, which is automatic for the integer-spectrum reason of Remark 5.5.

9 Related work and scope

Connes–Rovelli thermal time hypothesis. The closest published analog of the present construction is the Connes–Rovelli thermal time hypothesis [9], which identifies the Tomita modular flow of a general-covariant von Neumann algebra with the physical-time evolution. The present paper writes the Tomita machinery out on a finite-dimensional Type $\mathrm{I}_{\dim_W}$ factor obtained as the wedge-restricted Connes–van Suijlekom [10] truncation of the round S^3 Camporesi–Higuchi spectral triple, for a state that is itself defined from the chosen generator. The Connes–Rovelli construction starts from a state and reads off its flow; the present construction starts from the flow and builds the state, so it illustrates the machinery rather than testing the hypothesis.

Bisognano–Wichmann tradition and continuum modular Hamiltonians. The original Bisognano–Wichmann papers [1, 2] establish the modular-flow-as-boost identification for the Rindler wedge of Minkowski spacetime, with the analytic period $\sigma_{i\cdot 2\pi} = \mathrm{id}$ as a corollary of the Tomita–Takesaki KMS condition [26, 27] at $\beta = 2\pi$ in rapidity units. The continuum entanglement-Hamiltonian construction for a half-space in free CFT was developed by Casini and Huerta [5], and extended to spherical regions of S^d in CFT by Casini–Huerta–Myers [6]. The present construction is a formal analog of the Casini–Huerta–Myers spherical-region modular Hamiltonian on the truncated Camporesi–Higuchi triple over the round S^3, with the integer spectrum of the posited K_α^W standing in for the canonical 2π-period of the continuum modular flow; unlike Casini–Huerta–Myers, it does not derive the modular Hamiltonian from a state.

Lattice realisations. Zhang *et al.* [32] verified the Bisognano–Wichmann form of the modular Hamiltonian on critical quantum spin chains, finding the continuum BW prediction to hold to good approximation on finite lattices. The present paper differs in three respects: (i) the underlying object is a continuum spectral triple on a non-flat compact S^3 background, not a lattice discretisation of physical space; (ii) the finite-dimensional structure comes from the spectral truncation of D_{CH} via the Connes–van Suijlekom framework, not from a real-space discretisation; (iii) most importantly, Zhang et al. *extract* the modular Hamiltonian from a ground state and compare it with the BW form, which is a genuine test, whereas the present paper posits the generator, so its period closure is automatic. The analog of their test on the truncated triple is the [OPEN] question O0.

Algebraic QFT on curved backgrounds. The standard algebraic-QFT references for modular structures on Riemannian backgrounds are Strohmaier [25] (Reeh–Schlieder on stationary spacetimes) and Verch [31] (generally covariant spin-statistics and modular structure on curved backgrounds). These address the modular-theoretic content of axiomatic QFT on curved backgrounds. The present construction is spectral-triple-internal: it does not invoke a Wightman field algebra or a nuclearity argument, but rather works directly with the $\mathcal{O}_{n_{\max}} \subset B(\mathcal{H}_{n_{\max}})$ truncation of the $C^\infty(S^3) \subset B(L^2(S^3, \Sigma))$ continuum algebra and the associated Hilbert–Schmidt GNS construction.

Lorentzian signature and Krein-space spectral triples. The proper Lorentzian extension of the present construction would require a Krein-space spectral triple at Lorentzian signature (one time and three space dimensions), for which the canonical reference is Bizi–Brouder–Besnard 2018 [3]. The Bizi–Brouder–Besnard classification organises spectral triples of metric

signature (t, s) (t time and s space dimensions) by pairs $(m, n) = (t + s, t - s) \bmod 8$ (their p. 3; all BBB table, equation and page numbers in this paper follow arXiv:1611.07062 v2), where m alone is the metric dimension, and the underlying Connes axiom signs determine the Krein-space structure. The Riemannian Camporesi–Higuchi triple has $(t, s) = (0, 3)$ (three spatial directions), which gives $(m, n) = (3, -3 \bmod 8) = (3, 5)$ — this is BBB's formula extended formally to odd m (their tables cover even m, n only); Connes' KO-dimension 3 corresponds to it; the Lorentzian extension, one time and three space dimensions, has $(t, s) = (1, 3)$, which gives $(m, n) = (4, -2 \bmod 8) = (4, 6)$, their Table 3 entry $(4, 6)$. (Elsewhere in this paper a bare signature pair $(3, 0)$ or $(3, 1)$ is written in the order (s, t), space first; it is never a BBB (m, n) label.) The Bizi–Brouder–Besnard signs there (their Table 1 and eq. (3)) make J anticommute with the chirality, which leads to the present author's Sprint L0 prediction that the M3 sub-mechanism of the master Mellin engine trivialises at $(t, s) = (1, 3)$ on chirality-symmetric spectrum (`debug/lorentzian_l0_audit_memo.md` §3). The Krein-space lift and the replication of the present modular construction at $(t, s) = (1, 3)$ are carried out at finite cutoff in §11; the M3 prediction there turns out to be convention-dependent (§11.8). A continuum Lorentzian analog of the Latrémolière propinquity remains absent from the literature.

Twisted-spectral-triple emergence. The recent work of Nieuviarts [21–23] proposes a twisted-spectral-triple morphism that constructs a pseudo-Riemannian spectral triple from a Riemannian almost-commutative twisted spectral triple, presented as an algebraic alternative to Wick rotation. The construction does not apply to the present S^3 case because of the structural odd-dimensional restriction (§ 3.2). The Lorentzian-extension route for the present paper therefore runs through the direct Bizi–Brouder–Besnard 2018 Krein-space lift, not through a twist morphism.

Other NCG-side gaps the present paper does not close. A genuine Lorentzian propinquity (in the sense of an analog of the Latrémolière quantum Gromov–Hausdorff propinquity for Krein-space metric spectral triples) is open as of May 2026. Toyota [28], Farsi–Latrémolière [11, 12], and Rieffel [16] extend the quantum-metric convergence machinery (propinquity or state-space GH) in various directions (discrete groups of polynomial growth, collapse with abelian factors, analytic paths of Riemannian metrics, Berezin–Toeplitz on S^2), and Hekkelman–McDonald [15] construct a related noncommutative integral on truncations (not a convergence result), but none addresses Krein-space signature change. Constructing the Lorentzian propinquity is original NCG-mathematics work, not algorithmic.

10 Conclusion and open questions

We have set up, on the truncated Camporesi–Higuchi spectral triple $\mathcal{T}_{n_{\max}}$, a hemispheric wedge P_W aligned with the Hopf-base axis, a wedge Gibbs state built from the posited generator $K_\alpha^W = \mathrm{diag}(2m_j)$ on the wedge states (a relabelling, not a restriction of $2J_z$), and the corresponding GNS/Tomita construction. What holds is elementary and [SYMBOLIC]: the period closure $\sigma_{2\pi}(O) = O$ (Theorems 5.4, 6.3), automatic for any integer-spectrum generator and any O; the conjugacy $\sigma_t^{\mathrm{TT}}(O) = \sigma_{-t}^{\alpha}(O)$ (Theorem 7.1), Tomita's theorem for a Gibbs state built from K_α^W; and the agreement of the six witness instantiations (Corollary 8.1), which share one generator and differ only in the label κ_g. All are reproduced bit-exactly in floating point at $n_{\max} \in \{2, 3, 4, 5\}$. We also record the operator-level contrast between $2J_z$ ($e^{2\pi i K} = +I$) and the wedge Dirac ($e^{2\pi i D_W} = -I$).

The same holds on the signature-$(3, 1)$ Krein carrier of §11, where the period closure is again automatic and signature-blind; that section also records the Krein axioms and Krein-self-adjointness of the Lorentzian Dirac, the four Bizi–Brouder–Besnard-predicted sign relations

at $(4,6)$, the failure of their universal chirality anticommutation on the truthful Camporesi–Higuchi Dirac, and the chirality-parity Hilbert–Schmidt orthogonality of H_{local} and the wedge Dirac, whose residual carries $1/\pi^2$ only through the posited normalisation.

What is withdrawn (2026-09-28). An earlier version of this paper read these closures as lifting the four-witness Wick-rotation theorem from a structural correspondence to an identification at the operator-system level at every finite cutoff. That reading is withdrawn: the generator is posited rather than derived, the Tomita step is circular, and the witness agreement holds by construction, so nothing here identifies a Bisognano–Wichmann modular flow at finite cutoff. *What stands:* the continuum four-witness statement ($\beta = 2\pi/\kappa_g$) as prior art, and the elementary facts above.

Open questions. (O0) Derive the modular Hamiltonian from a Camporesi–Higuchi state [OPEN]. The falsifier the corpus named for this construction — compute K on $\mathcal{T}_{n_{\max}}$ as the modular Hamiltonian of a wedge-restricted Camporesi–Higuchi state (for instance the restriction to $P_W \mathcal{O}_{n_{\max}} P_W$ of a global state defined from D_{CH} alone), and only then test $\sigma_{2\pi} = \text{id}$ and compare K with $2J_z$ — has not been run. It is the finite-cutoff counterpart of the lattice test of Zhang *et al.* [32], and it is the question on which any finite-cutoff four-witness claim would rest.

(O1) Lorentzian extension at signature $(t,s) = (1,3)$ [finite-cutoff part carried out; continuum part descoped]. As drafted, O1 proposed to replicate the present operator-system construction on a Bizi–Brouder–Besnard Krein-space spectral triple at BBB $(m,n) = (4,6)$, to verify the Sprint L0 prediction that M3 trivialises but M1 stays under the BBB signs (their Table 1 and eq. (3)), and to produce a Lorentzian KMS–Tomita–Takesaki theorem inside the framework. The replication was carried out at finite cutoff (§11); the M3 prediction was falsified under the n_{fock}-parity convention (§11.8); the Krein period closures are automatic for the posited generator and yield no Lorentzian KMS–Tomita–Takesaki theorem; and the continuum-propinquity leg was descoped (Paper 45 [44]). The progress record below is kept as history.

Progress update (Sprints L2-B/C/D, 2026-05-16). The Hilbert-space and Lorentzian-Dirac levels of the $(3,1)$ extension have landed: Sprint L2-B (`geovac/krein_space_construction.py`) constructs the Krein space $\mathcal{K}_{n_{\max},N_t} = \mathcal{H}_{\text{GV}}^{n_{\max}} \otimes L^2(\mathbb{R}_t)_{\text{cutoff}}$ with fundamental symmetry $J = \gamma^0$ in the Peskin–Schroeder chiral basis, verifying all Krein axioms ($J^2 = +I$, $J^*J = I$, $J^* = J$, $\mathcal{K} = \mathcal{K}^+ \oplus \mathcal{K}^-$) bit-exactly at $(n_{\max}, N_t) \in \{1,2,3\} \times \{1,11,21\}$ and passing the load-bearing Riemannian-limit recovery at $N_t = 1$ bit-identically; Sprint L2-C (`geovac/lorentzian_dirac.py`) constructs $D_L = i \cdot [\gamma^0 \otimes \partial_t + D_{\text{GV}} \otimes I_{N_t}]$ via van den Dungen Proposition 4.1 with ∂_t centered FD plus Dirichlet zero BC (structurally forced for anti-Hermiticity and Krein-self-adjointness), verifying Krein-self-adjointness $D_L^{\times} = D_L$ and Riemannian-limit recovery $D_L|_{N_t=1} = iD_{\text{GV}}$ bit-exactly across the same panel. Sprint L2-D (`geovac/connes_axiom_audit_31.py`) verifies the four BBB-predicted sign relations at $(m,n) = (4,6)$ bit-exactly ($J_L^2 = +I$, $\{J_L, \gamma^5\} = 0$, $\{J_L, \eta\} = 0$, $J_L D_L = +D_L J_L$; Theorem 11.5), while the order-zero and order-one conditions fail at finite cutoff (Remark 11.6) and the universal axiom $\{\gamma^5, D_L\} = 0$ is a *structural finding*, failing on the truthful Camporesi–Higuchi Dirac (which is chirality-diagonal in the chiral basis and therefore commutes with γ^5; §11.2). This is structurally analogous to the Riemannian-side R3.5/present-paper trade between truthful CH (correct for axioms requiring $JD = +DJ$) and offdiag CH (correct for SDP-bounding Connes-distance constructions). The construction does *not* re-open WH1's commutative-factor convergence result (Paper 38 [40]); the Krein construction is an extension on top of the Riemannian foundation, not a re-test of it. An operator-system-level identification of the four-witness theorem at Lorentzian signature (which would first require the Riemannian [OPEN] question O0) was the named Sprint L2-E target, still open beyond sprint scale due to the absence of a published Lorentzian propinquity. O1 status remains OPEN *at the continuum / non-compact level*; the Hilbert-space and Dirac-operator levels are now closed at finite cutoff. On the *metric*-level finite-cutoff question (Paper 45 § open Q1) only the compactness half is established [SYMBOLIC]: the posited generator $K_\alpha = 2J_z$ is compact

(integer spectrum, $e^{2\pi iK} = I$). The further reading has two legs, priced separately. (a) That the finite-cutoff quantum metric is Euclidean, with the genuine reverse-triangle signature confined to the non-compact continuum limit, is [ARGUED] and *generator-independent*: it does not depend on which generator a wedge state selects. (b) That the Lorentzian signature is the Wick rotation of the KMS circle, a convention, is [ARGUED] and *conditional on O0*: it applies to the Bisognano–Wichmann generator only if K is derived from a wedge-restricted Camporesi–Higuchi state, which is open.

(O2) Non-tracial wedge states. The closure rests on the Gibbs structure $\rho_W = e^{-K_\alpha^W}/Z$ on a finite-dimensional Type $\mathrm{I}_{\dim_W}$ factor. For non-tracial wedge states (e.g., excited states above the BW vacuum, mixed states with non-Gibbs structure), the polar decomposition $S = J_{\mathrm{TT}}\Delta^{1/2}$ still goes through, but the integer-spectrum property of K_{TT} is not guaranteed, and the bit-exact closure may degrade to a soft propinquity-rate-controlled closure of the type Paper 38 supplies. The structural status of the non-tracial case is open.

(O3) Spectral-action Dirac vs. modular-Hamiltonian generator. Section 7.2 flagged that the framework's intrinsic Camporesi–Higuchi Dirac D_{CH} lifts the $\beta = 2\pi$ period to $-I$, whereas the posited K_α^W lifts it to $+I$. Which generator (if either) a Camporesi–Higuchi wedge state actually selects is O0; the present paper chooses K_α^W rather than deriving it. In general the spectral-action D_{CH} and the modular-Hamiltonian K_α^W may not even commute. The structural status of this distinction in extensions to other compact-Lie-group spectral triples (per Paper 40 [42]) is open and an interesting direction for follow-up.

Refinement (Sprint L2-E, 2026-05-16; § 11 below). The O3 question of signature-dependence is now *partially* closed: the $H_{\mathrm{local}} \neq D_W$ distinction is **signature-INDEPENDENT at the Riemannian limit** and refined upward at $N_t > 1$ by temporal-derivative content. At $N_t = 1$ on the Lorentzian Krein space, $\|H_{\mathrm{local}} - D_L^W\|_F$ equals the Riemannian-side residual $\|H_{\mathrm{local}} - D_{\mathrm{GV}}^W\|_F$ *bit-exact* at every tested $n_{\max} \in \{1, 2, 3\}$ (because $D_L = iD_{\mathrm{GV}}$ at $N_t = 1$ and H_{local} is real). This is a property of the posited construction, not evidence about the modular content of a Camporesi–Higuchi state. At $N_t > 1$ the Lorentzian Dirac $D_L = i(\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I)$ has temporal-derivative content that K_α^W/β cannot capture; the gap between the two operators widens with temporal cutoff. Which generator a Camporesi–Higuchi wedge state selects remains O0.

Further refinement (Hilbert–Schmidt orthogonality; §11.7). The signature-independent residual is Pythagorean: $\langle H_{\mathrm{local}}, D_W^L\rangle_{\mathrm{HS}} = 0$ as a chirality-parity identity (Theorem 11.14), and $\|H_{\mathrm{local}} - D_W^L\|_F^2 = \kappa_g^2 S(n_{\max})/(4\pi^2) + D(n_{\max})$ with S, D polynomials in $n_{\max}$ (Corollary 11.15). The sole transcendental, $1/\pi^2$, enters through the posited $\beta = 2\pi/\kappa_g$ — the continuum KMS/Bisognano–Wichmann period imported as a normalisation (a temporal-compactification 2π in the sense of Paper 35 [39]) — and is an input, not an output of the finite construction. The orthogonality is a statement about two chosen operators; it does not show that D_W is the "wrong" modular generator, which only O0 could decide. The S^3-specificity of the construction — the Bargmann–Segal S^5 Hardy sector lacks a spinor sector and a half-integer wedge — is the fourth layer of the Paper 24 § 5.3 [34] Coulomb/HO asymmetry.

(O4) Higher-rank compact non-abelian extensions. Paper 40 [42] extends the Paper 38 state-space Gromov–Hausdorff convergence theorem to compact connected semisimple Lie groups with bi-invariant metric (for general G conditionally on the per-group spin-window decomposition; Fejér-moment constant derived at rank 1; a circle factor carries the circle constant). The natural extension of the present paper would verify that the geometric BW-α generator analog on higher-rank groups (e.g., a relabelling built from a Cartan generator along the wedge-defining direction) still has integer spectrum on the spinor harmonic basis. Since any integer-weight Cartan generator closes automatically, that extension would be as elementary as the present one; the substantive higher-rank question is the analog of O0.

(O5) Cross-manifold modular structures. The Paper 24 [34] cross-manifold blocker $\mathcal{T}_{S^3} \otimes \mathcal{T}_{\mathrm{Hardy}(S^5)}$ remains unaddressed by the present paper. A cross-manifold modular construc-

tion (modular Hamiltonian on a tensor product of a round S^3 Camporesi–Higuchi triple and a holomorphic-sector S^5 Bargmann–Segal triple) would require an analog of Paper 39's state-space Gromov–Hausdorff tensor theorem [41] (Paper 39 does not claim Latrémolière propinquity) extended to mixed Riemannian / Hardy-sector spectral triples. This is also open.

Final remark on the M1 master Mellin engine. The period of the present construction is 2π because the posited generator has integer spectrum; that is the generic fact $e^{2\pi i n} = 1$, which says nothing about the Paper 38 Fejér-moment constant or about Bisognano–Wichmann. The M1 reading of the four-witness 2π remains at the metric-functional level (Paper 32 § VIII), where it was before.

11 Krein-carrier extension at finite cutoff

Provenance. This section is the successor of the former standalone Paper 43 (*Lorentzian extension of the four-witness Wick-rotation theorem on truncated $S^3 \times \mathbb{R}$ spectral triples at finite cutoff*), archived on 2026-09-28. Its surviving content — the Krein carrier and Lorentzian Dirac (§11.1), the axiom audit at Bizi–Brouder–Besnard signature $(4,6)$ (§11.2), the Lorentzian wedge (§11.3), the Krein-carrier period closure and the six-witness statement (§11.4), the Riemannian-limit recovery and the signature-independence of the H_{local} residual (§§11.5–11.6), the Hilbert–Schmidt orthogonality (§11.7) and the M3 convention finding (§11.8) — is collected here with its current tiers. Two headline readings of that paper are withdrawn and appear here only as history: (i) that the Krein-carrier period closure is an operator-system analog of the four-witness theorem at signature $(3,1)$ — it is the automatic integer-spectrum closure of Theorem 5.4, and signature-blind; (ii) that the $1/\pi^2$ of the Hilbert–Schmidt residual is a master-Mellin-engine M1 output — it is the square of the posited normalisation $\beta = 2\pi/\kappa_g$.

Open question (O1) of §10 named the extension to signature $(3,1)$ as the next step. Sprint L2 (May 2026) carries the Riemannian construction of §§2–8 onto a Krein carrier at finite cutoff, via the van den Dungen 2016 [30] Proposition 4.1 lift and the Bizi–Brouder–Besnard 2018 [3] classification. The construction is structurally additive: at $N_t = 1$ every object reduces bit-identically to its Riemannian counterpart (Proposition 11.10). Numerical statements below refer to the tested panel $n_{\max} \le 3$, $N_t \in \{1, 11, 21\}$ (supporting modules `geovac/krein_space_construction.py`, `geovac/lorentzian_dirac.py`, `geovac/connes_axiom_audit_31.py`, `geovac/modular_hamiltonian_lorentz` with the corresponding `tests/test_*.py` files).

11.1 The Krein carrier and the Lorentzian Dirac

Convention lock. We use the West-coast metric $\eta = \mathrm{diag}(+1,-1,-1,-1)$ on $S^3 \times \mathbb{R}$ and the Peskin–Schroeder chiral basis

$$\gamma^0 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}, \qquad \gamma^5 := i\gamma^0\gamma^1\gamma^2\gamma^3 = \mathrm{diag}(-I_2, +I_2). \tag{28}$$

The spatial space is the chirality-doubled Camporesi–Higuchi spinor space $\mathcal{H}_{\mathrm{GV}}^{n_{\max}}$ of dimension $\frac{2}{3} n_{\max}(n_{\max}+1)(n_{\max}+2) = 4, 16, 40$ at $n_{\max} = 1, 2, 3$, with basis labels $(n_{\mathrm{fock}}, l, m_j, \chi)$, $j = l + \frac{1}{2}$, $\chi = \pm 1$, on which the truthful Camporesi–Higuchi Dirac acts diagonally,

$$D_{\mathrm{GV}}\, \psi_{n_{\mathrm{fock}}, l, m_j, \chi} \;=\; \chi\, (n_{\mathrm{fock}} + \tfrac{1}{2})\, \psi_{n_{\mathrm{fock}}, l, m_j, \chi}. \tag{29}$$

The chirality label is identified with the γ^5 eigenvalue (up to a global sign): $\gamma^5_{\mathrm{spatial}} \psi_{\cdot,\chi} = -\chi\, \psi_{\cdot,\chi}$. The temporal slot is the bounded uniform grid $t_k = -T_{\max} + k \cdot 2T_{\max}/(N_t - 1)$, $k = 0, \ldots, N_t - 1$ (the singleton $\{0\}$ at $N_t = 1$). It is *not* compactified to S^1_β, which would create closed timelike curves on $S^3 \times S^1_\beta$ and conflict with the spacelike-reflection ingredient of the van den Dungen lift.

Definition 11.1 (Krein carrier). $\mathcal{K}_{n_{\max},N_t} := \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \otimes \mathbb{C}^{N_t}$, with fundamental symmetry $\mathcal{J} := \gamma^0_{\mathrm{spatial}} \otimes I_{N_t}$ (the chirality swap) and Krein form $\langle \psi, \phi \rangle_{\mathcal{K}} := \langle \psi, \mathcal{J}\phi \rangle$.

Proposition 11.2 (Krein axioms, [SYMBOLIC]). *$\mathcal{J}^2 = I$, $\mathcal{J}^* = \mathcal{J}$, and $\mathcal{K} = \mathcal{K}^+ \oplus \mathcal{K}^-$ orthogonally with $\dim \mathcal{K}^\pm = \dim \mathcal{K}/2$. In floating point all residuals are exactly 0 on the panel, because $\mathcal{J}$ is a real $\{0,1\}$ permutation matrix tensored with the identity.*

Definition 11.3 (Lorentzian Dirac).

$$D_L := i\big[\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I_{N_t}\big], \qquad (\partial_t)_{k,l} = \frac{\delta_{l,k+1} - \delta_{l,k-1}}{2\Delta_t}, \quad \Delta_t = \frac{2T_{\max}}{N_t - 1}, \tag{30}$$

with Dirichlet zero boundary conditions, so that ∂_t is real skew-symmetric. The prefactor $i^t = i$ is the $(s,t) = (3,1)$ instance of van den Dungen's Proposition 4.1.

Proposition 11.4 (Krein-self-adjointness, [SYMBOLIC]). $D_L^\times := \mathcal{J} D_L^\dagger \mathcal{J} = D_L$.

Proof. $D_L^\dagger = -i(-\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I)$, using $\partial_t^\dagger = -\partial_t$ and $D_{\mathrm{GV}}^\dagger = D_{\mathrm{GV}}$. Conjugating by $\mathcal{J}$ and using $(\gamma^0)^2 = I$ together with $\gamma^0 D_{\mathrm{GV}} \gamma^0 = -D_{\mathrm{GV}}$ (the chirality swap reverses the sign of the chirality-diagonal eigenvalue (29)) gives $i(\gamma^0 \otimes \partial_t + D_{\mathrm{GV}} \otimes I) = D_L$. The other prefactor $-i$ would give $D_L^\times = -D_L$. □

At $N_t = 1$ the grid is a point, $\partial_t = 0$, and

$$D_L\big|_{N_t=1} = i\, D_{\mathrm{GV}}, \tag{31}$$

the Riemannian Camporesi–Higuchi Dirac up to the global factor i that mediates between Krein- and Hilbert-self-adjointness. The chirality grading on $\mathcal{K}$ is $\gamma^5_{\mathcal{K}} := \gamma^5_{\mathrm{spatial}} \otimes I_{N_t}$.

11.2 Axiom audit at Bizi–Brouder–Besnard signature $(4,6)$

The Bizi–Brouder–Besnard classification of indefinite spectral triples by $(m,n) = (t+s, t-s) \bmod 8$ (their p. 3) places the Lorentzian signature with one time and three space dimensions, $(t,s) = (1,3)$, at $(m,n) = (4, -2 \bmod 8) = (4,6)$ (their Table 3 entry $(4,6)$), with primitive signs $\varepsilon = +1$, $\varepsilon'' = -1$, $\kappa = -1$, $\kappa'' = +1$ (their Table 1 and eq. (3)). The real structure is the lift of the Cl(3,1) charge conjugation $U_4 = i\gamma^2 = (i\sigma_y)_{\mathrm{chir}} \otimes (i\sigma^2)_{\mathrm{spin}}$,

$$J_L := U_L K, \qquad U_L := U_{L,\mathrm{spatial}} \otimes I_{N_t}, \qquad U_{L,\mathrm{spatial}}\, \psi_{n_{\mathrm{fock}},l,m_j,\chi} = \sigma_{\mathrm{chir}}(\chi)\, (-1)^{(2l+1-2m_j)/2}\, \psi_{n_{\mathrm{fock}},l,-m_j,-\chi}, \tag{32}$$

with K complex conjugation and $\sigma_{\mathrm{chir}}(\pm 1) = \mp 1$.

Theorem 11.5 (BBB-predicted-sign relations, [SYMBOLIC]). *On $\mathcal{K}_{n_{\max},N_t}$: (i) $J_L^2 = +I$; (ii) $\{J_L, \gamma^5_{\mathcal{K}}\} = 0$; (iii) $\{J_L, \mathcal{J}\} = 0$; (iv) $J_L D_L = D_L J_L$. All four residuals are exactly 0 on the panel.*

Proof. (i) $(i\sigma_y)^2 = -I_{\mathrm{chir}}$ and $(i\sigma^2)^2 = -I_{\mathrm{spin}}$, so $U_L^2 = +I$; U_L is real, so $J_L^2 = U_L \overline{U_L} = +I$. (ii),(iii) $i\sigma_y$ anticommutes with both γ^5 (diagonal $\mp$ in the chirality factor) and the chirality swap γ^0. (iv) γ^0, ∂_t and D_{GV} are real, so K flips the sign of the prefactor i; conjugation by U_L flips the sign of γ^0 (anticommutation) and of D_{GV} (U_L reverses χ in (29)). The three signs cancel: $J_L D_L J_L^{-1} = D_L$. □

Remark 11.6 (Order-zero and order-one conditions). The order-zero condition $[a, J_L b J_L^{-1}] = 0$ and the order-one condition $[[D_L, a], J_L b J_L^{-1}] = 0$, for multipliers a, b of the operator system, fail at finite cutoff, with Frobenius residuals up to $\sim 7\%$ and $\sim 11\%$ on a sample of three multipliers per panel cell. These are *not* finite-resolution artifacts expected to vanish: on the Riemannian side, Paper 32 [37] (§ IV) measures the order-zero failure *growing* with the cutoff over $n_{\max} = 2, 3, 4$ and, on a fixed generator set, approaching a nonzero limit. The cutoff dependence of the Lorentzian residuals has not been measured.

Proposition 11.7 (Universal anticommutation fails on truthful D_{GV}, [SYMBOLIC]). *The BBB universal relation $\{\chi, D\} = 0$ (their § 5(v), required at every signature) fails: $\{\gamma^5_{\mathcal{K}}, D_L\} = 2i\,(\gamma^5 D_{\mathrm{GV}}) \otimes I_{N_t} \neq 0$, with Frobenius norm $2\|D_{\mathrm{GV}}\|_F = 6.00,\ 18.33,\ 38.88$ at $N_t = 1$, $n_{\max} = 1, 2, 3$.*

Proof. $\{\gamma^5, \gamma^0\} = 0$ kills the temporal piece; γ^5 and D_{GV} are both chirality-diagonal, so they commute and $\{\gamma^5, D_{\mathrm{GV}}\} = 2\gamma^5 D_{\mathrm{GV}}$. □

A mutually inconsistent triple. (P1) chirality identified with the γ^5 eigenvalue; (P2) the Camporesi–Higuchi D_{GV} chirality-diagonal (29); (P3) the BBB universal axiom $\{\chi, D\} = 0$. Any two are compatible; (P1)+(P2) force $[\gamma^5, D_{\mathrm{GV}}] = 0$, contradicting (P3). Three resolutions exist: (R1) keep truthful D_{GV} and record the failure as scope; (R2) use the chirality-flipping off-diagonal variant $D_{\mathrm{GV}}^{\mathrm{offdiag}}$, which anticommutes with γ^5 but breaks bit-exact Riemannian-limit recovery; (R3) redefine γ^5 as an off-diagonal grading, abandoning (P1). This section adopts (R1): the construction is a Krein-self-adjoint pair with the four BBB-predicted-sign relations of Theorem 11.5, but not a full BBB indefinite spectral triple in the strict sense of their § 5(v). The finding is new at $(3, 1)$: the Riemannian triple has odd KO-dimension and no chirality axiom to fail. It does not affect the period closures below, which depend only on the generator $K_L^{\alpha,W}$, not on D_L.

11.3 Lorentzian wedge, Gibbs state and generator

Definition 11.8 (Lorentzian hemispheric wedge). $P_{W_L} := P_W^{\mathrm{spatial}} \otimes P_{t\geq 0}$, with P_W^{spatial} the hemispheric wedge of Definition 4.1 and $P_{t\geq 0}$ the diagonal projector on $\mathbb{C}^{N_t}$ selecting $t_k \geq 0$. Then P_{W_L} is a Hermitian idempotent, $\dim W_L = \frac{1}{2}\dim \mathcal{H}_{\mathrm{GV}}^{n_{\max}} \cdot |\{k : t_k \geq 0\}|$, and $P_{W_L}|_{N_t=1} = P_W^{\mathrm{spatial}}$.

The generator is $K_L^{\alpha} := K_\alpha \otimes I_{N_t}$ with K_α of Definition 5.1; its wedge operator $K_L^{\alpha,W} := K_\alpha^W \otimes I$ on W_L (the relabelling of Definition 5.1, not a compression) is diagonal with the positive odd integers two_m_j as eigenvalues. Under the Bisognano–Wichmann choice $H_{\mathrm{local}} := K_L^{\alpha,W}/\beta$ the wedge Gibbs state

$$\rho_W^L := e^{-\beta H_{\mathrm{local}}}/Z = e^{-K_L^{\alpha,W}}/Z \tag{33}$$

is β-independent. Its GNS Hilbert–Schmidt space is $M_{\dim W_L}(\mathbb{C})$, with modular operator $\Delta_L(X) = \rho_W^L X (\rho_W^L)^{-1}$, Tomita conjugation $J_L^{\mathrm{TT}}(X) = (\rho_W^L)^{1/2} X^* (\rho_W^L)^{-1/2}$ satisfying $(J_L^{\mathrm{TT}})^2 = +I$, and modular Hamiltonian $K_L^{\mathrm{TT}} := -\log \Delta_L$, whose spectrum is $\{n_j - n_k\} \subset \mathbb{Z}$ for $\{n_j\} = \mathrm{Spec}\, K_L^{\alpha,W}$. Three distinct antilinear operators are now in play and must not be conflated: J_{GV} on $\mathcal{H}_{\mathrm{GV}}$ ($J_{\mathrm{GV}}^2 = -I$), the Krein real structure J_L on $\mathcal{K}$ ($J_L^2 = +I$), and the state-dependent Tomita conjugation J_L^{TT} on the GNS space ($(J_L^{\mathrm{TT}})^2 = +I$).

11.4 Krein-carrier period closure at finite cutoff

Theorem 11.9 (Krein-carrier period closure, [SYMBOLIC]; automatic, as Theorem 5.4). *With $\sigma_t^{L,\alpha}(O) := e^{itK_L^{\alpha,W}} O e^{-itK_L^{\alpha,W}}$ and $\sigma_t^{L,\mathrm{TT}}(a) := \Delta_L^{it}(a)$ on $\mathcal{B}(\mathcal{K}_{W_L})$, for every $(n_{\max}, N_t)$:*

(a) $\sigma_{2\pi}^{L,\alpha}(O) = O$ *for every* O;

(b) $\sigma_{2\pi}^{L,\mathrm{TT}}(a) = a$ *for every* a;

(c) $\sigma_t^{L,\mathrm{TT}}(a) = \sigma_{-t}^{L,\alpha}(a)$ *for every* $t \in \mathbb{R}$;

(d) (six-witness statement, by construction.) *the six instantiations* (BW, $\mathrm{HH}_{M=1}$, $\mathrm{HH}_{M=2}$, $\mathrm{Sew}_{M=1}$, $\mathrm{Unruh}_{a=1}$, *yield identical* ρ_W^L, Δ_L *and* K_L^{TT}.

In floating point, the residuals of (a)–(c) sit at the 10^{-16}–10^{-15} *round-off level on the panel, and the cross-witness residual of (d) is exactly* 0.

Proof. $K_L^{\alpha,W}$ is diagonal with integer eigenvalues, so $e^{\pm 2\pi i K_L^{\alpha,W}} = I$, which gives (a). Since $\rho_W^L = e^{-K_L^{\alpha,W}}/Z$, $\Delta_L^{it}(a) = e^{-itK_L^{\alpha,W}}\, a\, e^{itK_L^{\alpha,W}}$ (the Z cancels), which is (c), and (b) follows from (a) and (c). For (d), the witnesses differ only in κ_g, which cancels in $\beta H_{\text{local}} = K_L^{\alpha,W}$. □

Every clause is elementary and none identifies a Bisognano–Wichmann flow. (a) and (b) hold for *any* integer-spectrum generator and *any* operator; (c) is Tomita's theorem for a Gibbs state built from the generator, so it is circular as a check on the generator; (d) holds because the six witnesses share one generator. The closure is also *signature-blind*: the posited generator is compact (integer spectrum, [SYMBOLIC]), so its flow is a $\beta = 2\pi$ circle rather than a non-compact boost, and on the argued reading [ARGUED] the $(3,1)$ label is a carrier choice the closure cannot distinguish from the Euclidean one (WH7; this compactness argument is separate from Paper 45's K^+ annihilation theorem [44], which concerns the Krein-positive compression). The former Paper 43 read (a)–(d) as an operator-system analog of the four-witness theorem at signature $(3,1)$; that reading is withdrawn (2026-09-28). The finite-cutoff test that would bear on it is the Riemannian O0 of §10.

11.5 Riemannian-limit recovery at $N_t = 1$

Proposition 11.10 (Riemannian-limit recovery, [SYMBOLIC])**.** *At* $N_t = 1$ *every construction of this section reduces bit-identically (Frobenius residual exactly* 0*) to its Riemannian counterpart:* $P_{W_L} \to P_W$ *(Definition 4.1),* $K_L^\alpha \to K_\alpha$ *(Definition 5.1),* $K_L^{\alpha,W} \to K_\alpha^W$, $\rho_W^L \to \rho_W$ *(§4.2),* $\Delta_L \to \Delta$ *and* $K_L^{\text{TT}} \to K_{\text{TT}}$ *(§6), and* $D_L \to iD_{\text{GV}}$ (31).

Proof. At $N_t = 1$, $\partial_t = 0$, $P_{t\geq 0} = I_1$, and every Kronecker product with I_1 is the identity operation. □

This is the falsifier that makes the construction a lift of the Riemannian one rather than a parallel construction; backed by `tests/test_modular_hamiltonian_lorentzian.py`. At $N_t > 1$ the temporal slot adds content (the temporal derivative in D_L, the larger wedge) invisible at $N_t = 1$.

11.6 Signature-independence of the H_{local} residual

Theorem 11.11 (H_{local} residual at the Riemannian limit, [SYMBOLIC])**.** *At* $N_t = 1$, $\|H_{\text{local}} - D_L^W\|_F = \|H_{\text{local}} - D_W^{\text{GV}}\|_F$, *with values* 2.1332, 6.5275, 13.854 *at* $n_{\max} = 1, 2, 3$ *for* $\kappa_g = 1$ *(*$\beta = 2\pi$*).*

Proof. $D_L^W|_{N_t=1} = iD_W^{\text{GV}}$ and H_{local} is real, so the cross term $2\,\text{Re}\,\text{Tr}(H_{\text{local}}\, iD_W^{\text{GV}})$ of the Lorentzian side vanishes (the trace of a product of two real matrices is real), and the Riemannian cross term $2\,\text{Tr}(H_{\text{local}} D_W^{\text{GV}})$ vanishes by Theorem 11.14 below. Both sides equal $\|H_{\text{local}}\|_F^2 + \|D_W^{\text{GV}}\|_F^2$. □

The §7.2 / O3 distinction between the wedge Dirac and the posited generator therefore persists verbatim at signature $(3,1)$ at the Riemannian limit, for the elementary reason in the proof. At $N_t > 1$ the Lorentzian residual grows with the temporal resolution [MEASURED; a PM measurement of 2026-09-28 at $n_{\max} = 1, 2, 3$, $N_t \leq 41$, $T_{\max} = 1$, outside the declared test panel and not test-backed]: at $\kappa_g = 1$ the finite-difference ∂_t term comes to dominate and the growth approaches $N_t^{3/2}$ (residual ratio ≈ 2.8 per doubling at $n_{\max} = 1$, lower at $n_{\max} = 2, 3$ where the spatial part is larger) — from the temporal-derivative content of D_L, which $K_L^{\alpha,W}/\beta$ cannot capture. This is a property of the posited construction; it does not show which generator a Camporesi–Higuchi wedge state selects (O0, open).

11.7 Hilbert–Schmidt orthogonality: a chirality-parity identity

Let Π_W be the wedge chirality grading ($\gamma^5_{\mathcal{K}}$ restricted to the wedge; $\Pi_W^2 = I$, $\Pi_W^\dagger = \Pi_W$, $\mathrm{Tr}\Pi_W = 0$).

Lemma 11.12 (Parity). *If Π is a unitary involution, $[\Pi, A] = 0$ and $\{\Pi, B\} = 0$, then $\mathrm{Tr}(A^\dagger B) = 0$.*

Proof. $\mathrm{Tr}(A^\dagger B) = \mathrm{Tr}(\Pi A^\dagger \Pi\, \Pi B \Pi) = -\mathrm{Tr}(A^\dagger B)$. □

Lemma 11.13 (Chirality pairing). *Let $\Pi = \mathrm{diag}(\chi_i)$ with ± 1 eigenspaces paired group-by-group, let A be diagonal with equal entries on each pair, and let $B = \Pi|B|$ with $|B|$ diagonal and equal on each pair. Then $\mathrm{Tr}(A^\dagger B) = 0$.*

Proof. Each pair contributes $\bar{a}\,(b_+ + b_-) = \bar{a}\,(|b| - |b|) = 0$. □

Theorem 11.14 (Hilbert–Schmidt orthogonality, [SYMBOLIC]). *For every $(n_{\max}, N_t)$, every rescaling of the generator (in particular every β and every witness κ_g),*

$$\langle H_{\mathrm{local}}, D_L^W \rangle_{\mathrm{HS}} := \mathrm{Tr}\big(H_{\mathrm{local}}^\dagger D_L^W\big) = 0.$$

Proof. $H_{\mathrm{local}} = K_L^{\alpha,W}/\beta$ is diagonal with entries two_m_j/β independent of χ, so it is Π_W-even and χ-independent. Split D_L^W into its temporal part $i\gamma^0 \otimes \partial_t$, which is Π_W-odd because $\{\gamma^5, \gamma^0\} = 0$ (Lemma 11.12), and its spatial part $iD_{\mathrm{GV}}^W \otimes I$, which is Π_W-*even* but sign-paired, $D_{\mathrm{GV}}^W = \Pi_W |D_{\mathrm{GV}}^W|$ with $|D_{\mathrm{GV}}^W| = n_{\mathrm{fock}} + \frac{1}{2}$ independent of χ (29) (Lemma 11.13). Both traces vanish. □

The spatial factor is not Π_W-odd: it commutes with Π_W, which is precisely the failure of the BBB universal axiom (Proposition 11.7). The orthogonality is a chirality-parity identity about two chosen operators; it is independent of β and does not decide which generator a wedge state selects.

Corollary 11.15 (Pythagorean residual). *At $N_t = 1$, with $\beta = 2\pi/\kappa_g$,*

$$\|H_{\mathrm{local}} - D_L^W\|_F^2 = \frac{\kappa_g^2\, S(n_{\max})}{4\pi^2} + D(n_{\max}), \quad S = \frac{n_{\max}(n_{\max}+1)(n_{\max}+2)(2n_{\max}^2 + 4n_{\max} - 1)}{15}, \quad D = \frac{n_{\max}(n} \tag{34}$$

The corollary follows from Theorem 11.14 by expanding the norm: $S = \|K_L^{\alpha,W}\|_F^2 = 4\sum_{\mathrm{wedge}} m_j^2$ is the cumulative wedge angular-momentum trace and $D = \|D_L^W\|_F^2 = \sum_{\mathrm{wedge}} (n_{\mathrm{fock}} + \frac{1}{2})^2$ the squared-Dirac trace. The two polynomial closed forms are [MEASURED]: in the original measurement they were identified at $n_{\max} = 1, \ldots, 6$ (PSLQ at 100 digits on $\{r^2, 1, 1/\pi^2\}$), and the orthogonality was measured across a larger panel and across the six witnesses at $n_{\max} \le 3$ (max 8.9×10^{-16}); neither the PSLQ step nor the six-witness check is in the regression suite. The backing test `tests/test_paper43_pythagorean_hs_orthogonality.py` recomputes the closed forms from the live wedge operators at $n_{\max} = 1, \ldots, 4$ ($\kappa_g = 1$) and checks the orthogonality on 7 cells (Riemannian $n_{\max} = 2, 3, 4$; Lorentzian $(n_{\max}, N_t) = (2,1), (3,1), (2,3), (3,3)$, tolerance 10^{-12}). The decomposition persists at $N_t > 1$, the temporal-derivative contribution adding only to the rational part.

Transcendental tag. The sole transcendental is $1/\pi^2$, and it enters only through the posited normalisation $\beta = 2\pi/\kappa_g$ in $H_{\mathrm{local}} = K_\alpha^W/\beta$, squared by the Frobenius norm. In Paper 18/Paper 34 [38] terms it is the 2π of a thermal (temporal-compactification) circle (Paper 35 [39]) imported as input — the continuum Bisognano–Wichmann/KMS period — not an output of any spectral computation on the truncated triple. The residual lies in $\mathbb{Q}[n_{\max}] \oplus \pi^{-2}\mathbb{Q}[n_{\max}]$,

the algebraic class of the M1 sector of the master Mellin engine (Paper 32 § VIII), but no Mellin mechanism producing it is exhibited, and it is not the M1 content of the Paper 38 Fejér-moment constant. The former Paper 43 read it as the squared volume factor $1/\mathrm{Vol}(S^1)^2$ and a $k = 0$ instance of the case-exhaustion theorem; withdrawn (2026-09-28).

11.8 The M3 sub-mechanism is sectional in n_{fock}-parity

The Sprint L0 prediction (§3.1) that the M3 sub-mechanism of the master Mellin engine trivialises at $(3, 1)$ on chirality-symmetric spectrum is convention-dependent. Under the n_{fock}-parity convention of the Paper 28 [36] vertex-parity sum, $\sum_{n=1}^{N} g_n(-1)^n/(n+\frac{1}{2})^4$ with $g_n = 2n(n+1)$ counting both chiralities, the value at $(3, 1)$ is bit-identical to the Riemannian one at $n_{\max} \leq 5$, so M3 does not trivialise. Under a chirality-pairing convention ($\chi = +1$ even, $\chi = -1$ odd) the difference vanishes tautologically, but that convention does not access the Catalan / Dirichlet-β content of Paper 28. The M3 sub-mechanism is therefore sectional in n_{fock}-parity, not in spacetime signature; the case-exhaustion classification of Paper 32 § VIII is unchanged.

11.9 Honest scope at finite cutoff

- Everything in this section is a finite-$(n_{\max}, N_t)$ statement. No continuum limit is taken. A Lorentzian analog of the Latrémolière propinquity [17, 18] is absent from the literature (Toyota [28], Farsi–Latrémolière [11, 12], Hekkelman–McDonald [15] are Riemannian), and the framework's attempt was descoped on 2026-06-09 (Paper 45's K^+ annihilation theorem: the natural Krein-positive compression fails structurally). Separately, the posited generator is compact [SYMBOLIC]; that no Lorentzian metric statement is therefore available at finite cutoff for it is [ARGUED] (Paper 45 Q1). The Nieuviarts twist morphism [22, 23] is restricted to even dimension and does not apply to S^3.
- The construction uses truthful D_{GV} (resolution R1 of §11.2), so it is not a full BBB indefinite spectral triple; the period closures are D_L-independent and unaffected.
- The cross-manifold product with the Bargmann–Segal S^5 Hardy sector (Paper 24 [34] § V) and the inner-factor calibration question are not addressed.
- The operator-system substrate on the full Krein space (the propagation number and its envelope dependence) is Paper 44 [43], which takes the carrier of §11.1 as input.

A Computational verification protocol

The numerical verification of the period closures of Theorems 5.4 and 6.3 and of the cross-witness collapse of Table 3 is implemented in the supporting Python module `geovac/modular_hamiltonian.py` of the present author's framework codebase (∼1560 lines), with test coverage in `tests/test_modular_hamiltoni` (∼1070 lines, 70 tests: 66 fast plus 4 slow). The structure is briefly described here.

A.1 Module API

The principal classes are:

- `HemisphericWedge(axis="hopf")`: implements the projection P_W of Definition 4.1 via the equatorial-reflection involution R_{polar} on the full-Dirac spinor basis.
- `TomitaConjugation(U)`: implements the antilinear J_{TT} via a stored unitary U with action $J_{\mathrm{TT}}(\psi) = U \cdot \overline{\psi}$; exposes `verify_J_squared_positive_identity` and `verify_J_squared_negative_iden` as explicit sanity checks distinguishing J_{TT} from J_{GV}.

- `KMSState(beta, H)`: implements the wedge-restricted Gibbs state $\omega_{\beta,H}$ of (10), including the canonical analytic-continuation verification of the KMS condition at the expectation level.
- `ModularHamiltonian(wedge, basis, D, kappa_g)`: implements the BW-α construction with $K = K_\alpha^W$, the real-time flow σ_t^α, the period-closure verifier `verify_modular_periodicity`, and accessors to the Tomita–Takesaki structure (`tomita_structure`, `k_alpha_geometric`, `k_tomita`, `compare_alpha_vs_tomita`).
- `TomitaModularStructure(rho)`: implements the BW-γ Tomita construction on the GNS Hilbert–Schmidt space, with $\Delta = (\rho^{-1})^T \otimes \rho$, $K_{\mathrm{TT}} = -\log\Delta$, and the modular flow $\sigma_t^{\mathrm{TT}}(a) = \rho^{it} a \rho^{-it}$ on the algebra.

Witness factories `for_bisognano_wichmann`, `for_hartle_hawking`, `for_sewell`, `for_unruh` instantiate `ModularHamiltonian` with witness-specific κ_g values: BW witness ($\kappa_g = 1$), HH ($\kappa_g = 1/(4M)$), Unruh ($\kappa_g = a$). Each factory returns a `ModularHamiltonian` object whose period-closure verifier returns the label `STRONG_IDENTIFICATION` at every tested $n_{\max}$; the label records the automatic closure, not an identification (Remark 5.5). The witness factories share the generator K_α^W and differ only in κ_g, so their agreement is by construction.

A.2 Test coverage

The test suite `tests/test_modular_hamiltonian.py` verifies the following: wedge projector idempotency and Hermiticity ($P_W^2 = P_W$, $P_W^* = P_W$); Tomita conjugation properties ($J_{\mathrm{TT}}^2 = +I$, $J_{\mathrm{TT}}^2 \neq -I$, antilinearity, operator-action consistency); KMS state normalisation, partition-function positivity, real expectation on Hermitian operators; modular periodicity $\sigma_{2\pi}(O) = O$ at $n_{\max} \in \{2, 3, 4\}$ for the first five non-identity multipliers in `FullDiracTruncatedOperatorSystem`; cross-witness collapse at the operator-action level; KMS expectation-level condition for sample operator pairs (A, B); state-space GH bound cross-check (residual $\ll \gamma_{n_{\max}}$); operator-system non-leakage ($\sigma_{2\pi}(M)$ stays in $\mathcal{O}_{n_{\max}}$); modular flow group law $\sigma_t \cdot \sigma_s = \sigma_{t+s}$ and identity at $t = 0$. The slow tests verify the closure at $n_{\max} = 5$ (∼80 seconds wall time per slow test).

A.3 Wall time and reproducibility

Per-$n_{\max}$ wall time on a standard laptop (Intel i7, Python 3.11, NumPy 1.26): $\sim$ 0.4 s at $n_{\max} = 2$, $\sim$ 3.5 s at $n_{\max} = 3$, $\sim$ 45 s at $n_{\max} = 4$, $\sim$ 260 s at $n_{\max} = 5$ for the BW-α construction; $\sim$ 0.2 s, $\sim$ 3.7 s, $\sim$ 45 s, $\sim$ 545 s for the BW-γ construction (the $n_{\max} = 5$ cost dominated by the 4900×4900 GNS-Hilbert–Schmidt diagonalisation). Beyond $n_{\max} = 5$ the GNS construction becomes expensive; the structural reading of the closure is unchanged at higher $n_{\max}$ since the integer-spectrum mechanism holds at any cutoff. Computational drivers and structured output JSON are available in the present author's framework documentation under `debug/l1_modular_hamiltonian_compute.py` and `debug/l1_tighten_tomita_compute.py` respectively.

B Cross-references to the GeoVac framework

The present paper is part of an integrated NCG-research programme developed by the present author. We briefly note the principal cross-references for the reader's orientation.

Paper 38 ([40], state-space GH convergence on S^3). The convergence of the truncated Camporesi–Higuchi spectral triples $\mathcal{T}_{n_{\max}}$ to the continuum $\mathcal{T}_{S^3}$ in van Suijlekom's state-space Gromov–Hausdorff distance is established in Paper 38 with explicit rate (7). The L1' / L2 / L3 / L4 lemmas of Paper 38 supply the metric-side analytic infrastructure that the present paper

builds on; Paper 38's fifth lemma (the propinquity assembly) was withdrawn on 2026-09-03 and its convergence theorem is carried by the lifted-state construction, which the present paper does not depend on either way. The Fejér-moment constant $4/\pi = \mathrm{Vol}(S^2)/\pi^2$ of that bound is read (a reading, not a derivation; Paper 38) as carrying the $k = 0$ M1 volume content; the period closure of the present Theorems 5.4–6.3 is driven instead by the integer spectrum of the posited generator, and no link between the two is claimed.

Paper 40 ([42], universal $4/\pi$ on compact simple Lie groups). The Paper 38 state-space Gromov–Hausdorff convergence extends to compact connected semisimple Lie groups with bi-invariant metric (for general G conditionally on the per-group spin-window decomposition; a circle factor carries the circle constant), with the Fejér-moment constant $4/\pi$ (dual-Coxeter normalisation of the SU(2) metric; $2/\pi$ on the unit round S^3, the constant scaling with the metric — Paper 38) reported universal across the simple groups of that class (for semisimple groups with several simple factors the constant is open) via the Plancherel-weight × Vandermonde-Jacobian cancellation (Paper 40 § 3.2; derived at rank 1). The modular structure of the present paper plausibly extends to the higher-rank cases (Section 10 O4); the integer spectrum of the posited wedge generator K_α^W on the spinor harmonic basis is a property of SU(2) (where the half-integer weights double to integers), and the higher-rank analog requires verifying that the relevant Cartan generator along the wedge axis still has the right spectral property.

Paper 32 §VIII (master Mellin engine case-exhaustion theorem). The case-exhaustion theorem [37, Thm. thm:pi_source_case_exhaustion] of Paper 32 §VIII (in the present author's framework documentation) states (**[CONDITIONAL]**, first fifteen Paper 34 projections) that every π in such a chain engages one of three master Mellin engine sub-mechanisms: M1 (volume slot, $\mathrm{Vol}(S^2)/4 = \pi$, at operator order $k = 0$), M2 (Seeley–DeWitt $\sqrt{\pi} \times \bigoplus_k \pi^{2k}\,\mathbb{Q}$ at $k = 2$), or M3 (vertex-parity Hurwitz / Catalan G / Dirichlet L at $k = 1$). The 2π in the modular period of the present paper comes from the integer spectrum of the posited generator ($e^{2\pi i n} = 1$) and is not shown here to be the M1 signature. The Bisognano–Wichmann reading of M1 on temporal compactifications ([37, Rem. rem:bisognano_wichmann_reading]) remains a structural correspondence at the metric-functional level; the present paper does not lift it (an earlier reading that it did is withdrawn, 2026-09-28).

Paper 34 §III.27 (Wick rotation projection). The Wick rotation projection in the present author's projection taxonomy [38] is the structural mechanism through which the framework's Euclidean spectral content is read as a Lorentzian physical observable. The Sprint L0 audit (`debug/lorentzian_l0_audit_memo.md`) classifies this projection in the WICK_MAP_FREE bucket at the metric-functional level. The present paper does not change that status: its finite-cutoff period closure is automatic for the posited generator, so the framework still relies on the published Wick-rotation chain for the four-witness period (withdrawn reading, 2026-09-28).

Paper 24 ([34], Coulomb / HO asymmetry). The cross-manifold blocker named in Paper 24 §V (Riemannian S^3 ⊗ holomorphic-sector S^5 Bargmann–Segal triple) carries through to the modular construction: the present paper does not address the cross-manifold modular structure, and the natural extension would require a mixed Riemannian / Hardy-sector tensor-product propinquity that does not yet exist in the literature.

Paper 2 ([33], fine-structure-constant observation). The present paper does not impinge on Paper 2's status as an Observation. The $n_{\max} = 3$ cutoff of Table 1 happens to coincide with Paper 2's selection-principle truncation ($\dim \mathcal{H}_3 = 4 + 12 + 24 = 40$, the cumulative node count, numerically equal to Paper 2's single-level degeneracy $g_3^{\mathrm{Dirac}} = \Delta^{-1} = 40$ — different objects that agree only at $n_{\max} = 3$); this is a numerical coincidence at the Hilbert-space dimension

level, not a derivation of Paper 2's combination rule $K = \pi(B + F - \Delta)$. The combination rule remains a numerical observation without first-principles derivation.

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